

SCADA-Based Investigation of Early Blade Structural Damage Detection

Turbine 14 Case Study

Technical investigation report

August 9, 2026

Abstract

This report documents an investigation into whether blade structural damage can be detected early using SCADA data and stoppage or maintenance information. The selected case is Turbine 14, where blade structural damage involving the tip, leading edge, and trailing edge was detected in November 2023. Turbines 12, 13, 14, and 15 are analyzed as a local peer group. Confidential layout material and the initial confidential blade-damage event list have been removed from this version. The report follows the development of the analysis from the power curve, to global C_p -lambda curves, to pitch-conditioned C_p -lambda analysis. The final conclusion is that pitch angle is essential for interpreting the C_p -lambda relationship, and the most reliable pitch bands do not show a clear collapse in maximum aerodynamic efficiency for Turbine 14. A narrowing of the high-efficiency lambda range is observed through Delta Lambda 90 in the two most reliable pitch bands (1-2 and 2-3 degrees); tracked as a multi-year trend rather than a single-year snapshot, this metric declines steadily into the November 2023 detection date and recovers afterward, once the blade was repaired, providing usable evidence that blade structural damage can be detected through this metric ahead of severe failure. The report therefore recommends a future classification AI trained by pitch-angle range, incorporating multi-year Delta Lambda 90 trend as an input feature, and validated using known blade, pitch-system, generator, and data-quality cases.

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1 Problem Identification and Case Selection

1.1 Problem Definition

The objective of this investigation is to determine whether blade structural damage in a wind turbine can be detected early, before the damage becomes severe or visually significant, using only SCADA information and stoppage or maintenance records. SCADA data do not directly measure blade condition. Instead, the data describe turbine operation, including generated power, wind speed, rotor behavior, pitch operation, turbine status, and derived aerodynamic quantities. Therefore, the investigation attempts to detect blade structural damage indirectly through changes in aerodynamic or operational behavior.

The practical motivation is that blade structural damage can develop progressively. If the damage can be detected earlier using existing operational data, inspections and maintenance can be scheduled before the damage becomes critical, reducing downtime and reducing the risk of secondary damage.

1.2 Selected Case

The selected case for this investigation is Turbine 14, where blade structural damage involving the tip, leading edge, and trailing edge was detected in November 2023. The purpose of this report is not to reproduce confidential event lists, but to use this known case as a reference point for evaluating whether SCADA data and stoppage information could have shown earlier aerodynamic or operational symptoms.

1.3 Turbines Analyzed and Peer-Comparison Rationale

The analysis includes Turbines 12, 13, 14, and 15. Turbine 14 is the target turbine. Turbines 12, 13, and 15 are included as peer turbines. Peer comparison matters because turbines in the same local operating environment are expected to experience broadly similar wind and terrain conditions. If Turbine 14 deviates from the peer group while the peer turbines remain stable, the deviation is more likely to be turbine-specific rather than caused by general site conditions.

Turbine 13 is especially important because it had almost no major problems throughout the analyzed years and is therefore treated as the most stable reference turbine.

1.4 Known Operational Context Used for Interpretation

The SCADA results cannot be interpreted without maintenance and stoppage context. The known context used in this investigation is summarized below.

Turbine	Known context	Importance for interpretation
T12	Pitch-system problems in 2021 and 2023.	T12 cannot be assumed to be perfectly healthy in every year. Some pitch-band behavior may reflect pitch-system conditions.
T13	Almost no major problems throughout the analyzed period.	T13 is the most stable reference turbine.
T14	Pitch-system problem in 2020, and blade structural damage detected in November 2023.	T14 is the target turbine, but earlier pitch history must be considered when interpreting long-term trends.
T15	Catastrophic blade problem in 2020, including blade breakage, and hydraulic pitch-system problem in 2021.	T15 demonstrates that serious faults can still have misleadingly high apparent C_p - λ performance.

Table 1: Known operational and maintenance context used for interpretation.

2 Available Data, Betz Limit, and Power Curve Basis

2.1 Available SCADA Variables at the First Stage

The first stage uses the two most direct SCADA variables: generated power and wind speed. Generated power represents the electrical power produced by the turbine. Wind speed is used as the reference wind input for the turbine. These two variables allow the construction of a power curve, where generated power is plotted as a function of wind speed.

2.2 Theoretical Basis: Betz Limit and Power in the Wind

The aerodynamic starting point is that a wind turbine extracts kinetic energy from moving air. The power available in the wind is proportional to the cube of wind speed:

$$P_{wind} = \frac{1}{2}\rho AU^3, \quad (1)$$

where ρ is air density, A is rotor swept area, and U is wind speed. The Betz limit states that the maximum theoretical wind turbine power coefficient is $16/27 \approx 0.593$; no ideal open-flow turbine can extract more than this fraction of the kinetic power in the wind. This theoretical limit arises from actuator disk momentum theory: if all kinetic energy were extracted, air would stop behind the turbine and continuous flow would not be possible. Public technical references present the key relation $P = 0.5\rho C_p A v^3$ and the Betz limit $C_{p,max} = 16/27 \approx 0.593$ [1, 2].

The actual turbine output can therefore be written in simplified form as:

$$P_{out} = \frac{1}{2}\rho AU^3 C_p. \quad (2)$$

This equation forms the bridge between generated power, wind speed, and the turbine power coefficient.

2.3 Measured Wind Speed Is an Approximation

The theoretical wind speed in actuator disk theory is the undisturbed upstream free-stream wind speed, often denoted U_1 . In practice, the SCADA wind-speed measurement is taken at or near the turbine nacelle or rotor region, not at the ideal far-upstream location. Therefore, the measured wind speed is affected by the rotor, nacelle position, local flow distortion, turbulence, wake effects, and sensor location.

This is important because wind power varies with U^3 . A small wind-speed error can produce a large difference in estimated available power. Therefore, the SCADA wind speed is treated as a practical operational approximation, not an exact theoretical free-stream wind speed. The power curve is useful for comparison, but it is not a perfect physical representation of the ideal wind power assumed in Betz theory.

2.4 Power Curve as First Diagnostic Tool

The power curve plots generated power against wind speed. If a blade has structural damage, the turbine might produce less power for the same wind speed. However, the power curve is also affected by pitch control, curtailment, turbulence, air density, and turbine operating mode. Therefore, the power curve can detect large production changes, but it may not detect subtle or compensated faults.

3 Power Curve Limitation

3.1 Turbine 14 Versus Turbine 13 in MARCHA Operation

Power curves were first examined using MARCHA operation, meaning periods when the turbines were running normally. Turbine 14 was compared with Turbine 13 because Turbine 13 is considered a stable blade-failure-free reference turbine. In 2023, Turbine 14 and Turbine 13 showed visually similar power curves. Both had the expected increasing power with wind speed and rated-power saturation near 2 MW.

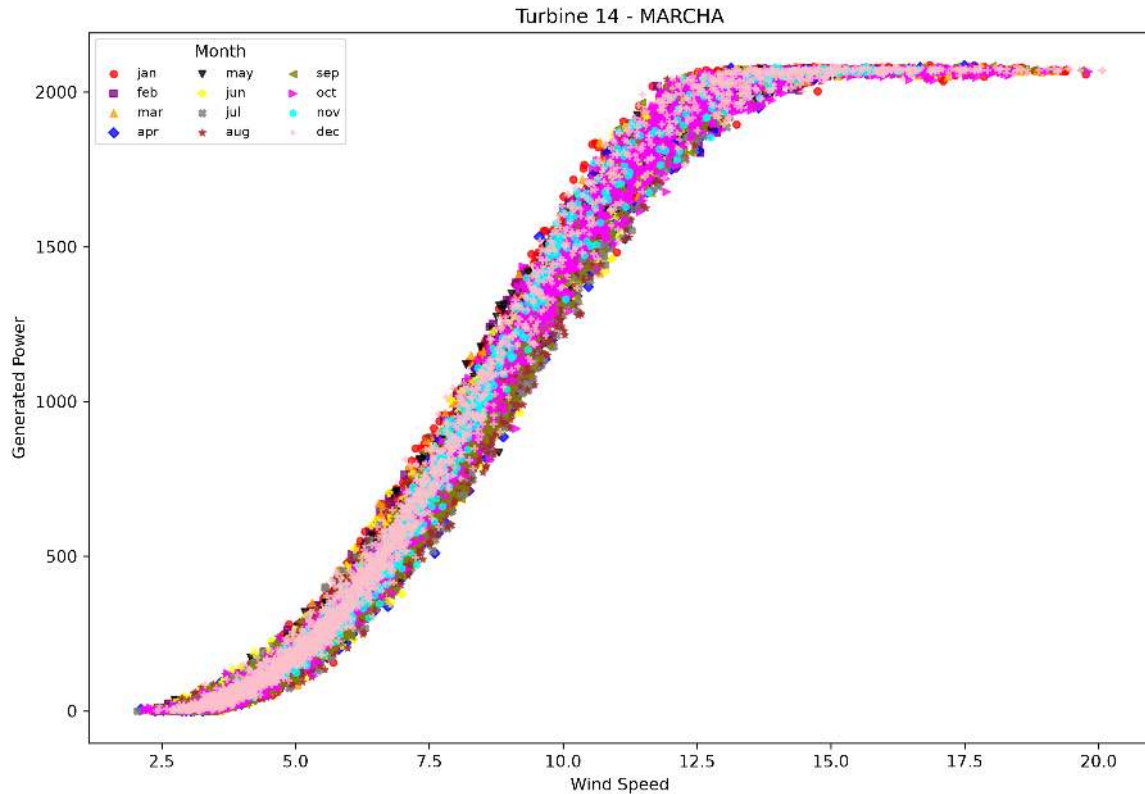


Figure 1: Turbine 14 MARCHA power curve by month. The curve appears broadly normal despite the known blade structural damage detection in November 2023.

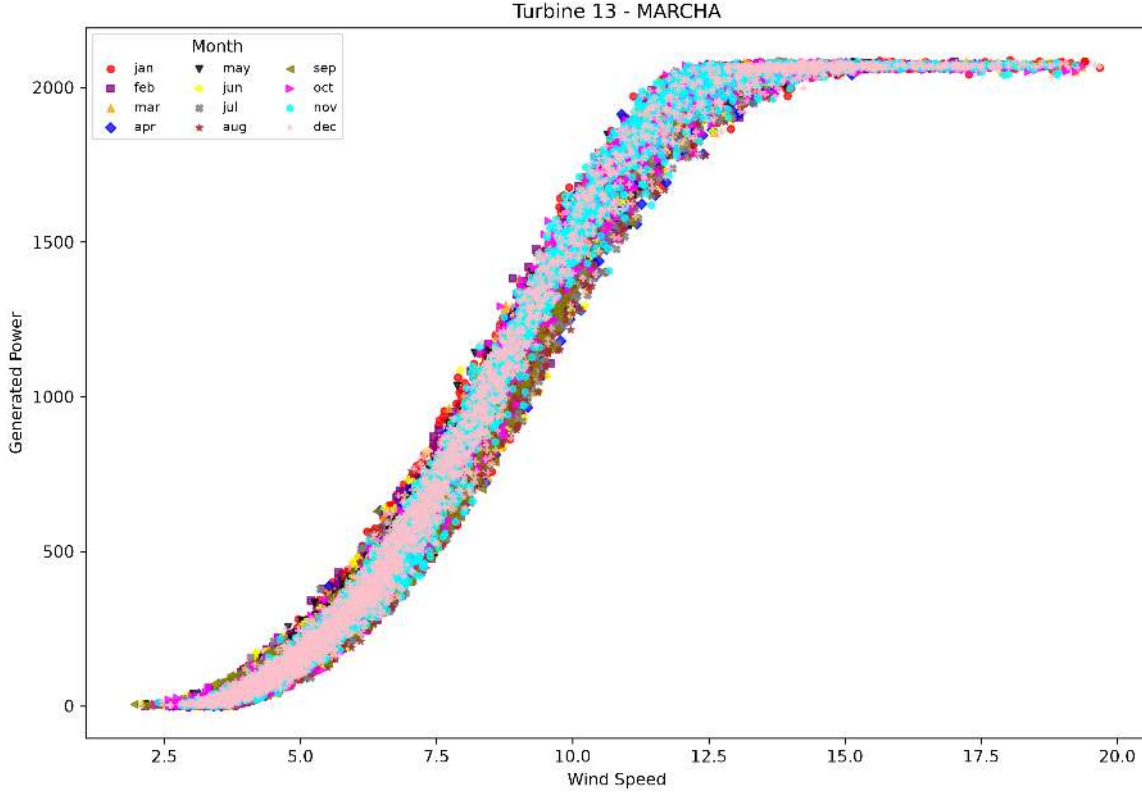


Figure 2: Turbine 13 MARCHA power curve by month. Turbine 13 is treated as the most stable reference turbine.

The power curve comparison showed no obvious distinction between the known damaged turbine and the stable reference turbine. This result was also observed in other turbines and another wind farm: turbines with known faults can still show apparently normal power-curve behavior during normal running periods. Therefore, the power curve alone cannot infer blade structural damage, pitch-system damage, or generator damage.

The key conclusion from this stage is that a normal power curve does not prove turbine health.

4 From Power Curve to Global C_p -Lambda Analysis

4.1 Need to Include Rotor Rotation

The power curve uses only generated power and wind speed. To study the aerodynamics more deeply, rotor rotation must also be considered. The available SCADA information includes rotor speed or RPM, which allows the calculation or use of tip-speed ratio.

The tip-speed ratio is:

$$\lambda = \frac{\Omega R}{U}, \quad (3)$$

where Ω is rotor angular speed, R is rotor radius, and U is wind speed. Rotor angular speed can be calculated from RPM using:

$$\Omega = \frac{2\pi \cdot RPM}{60}. \quad (4)$$

The power coefficient is:

$$C_p = \frac{P_{out}}{\frac{1}{2}\rho A U^3}. \quad (5)$$

Combining C_p and λ gives the C_p -lambda curve, intended to represent aerodynamic performance:

$$C_p = f(\lambda). \quad (6)$$

4.2 Global C_p -Lambda Results

The global C_p -lambda plots for Turbines 12, 13, and 14 showed that Turbine 14 did not stand out as faulty. In some metrics, Turbine 14 even appeared similar to or better than the stable reference turbine. This means global C_p -lambda analysis, while more aerodynamic than the power curve, still did not detect the known blade structural damage.

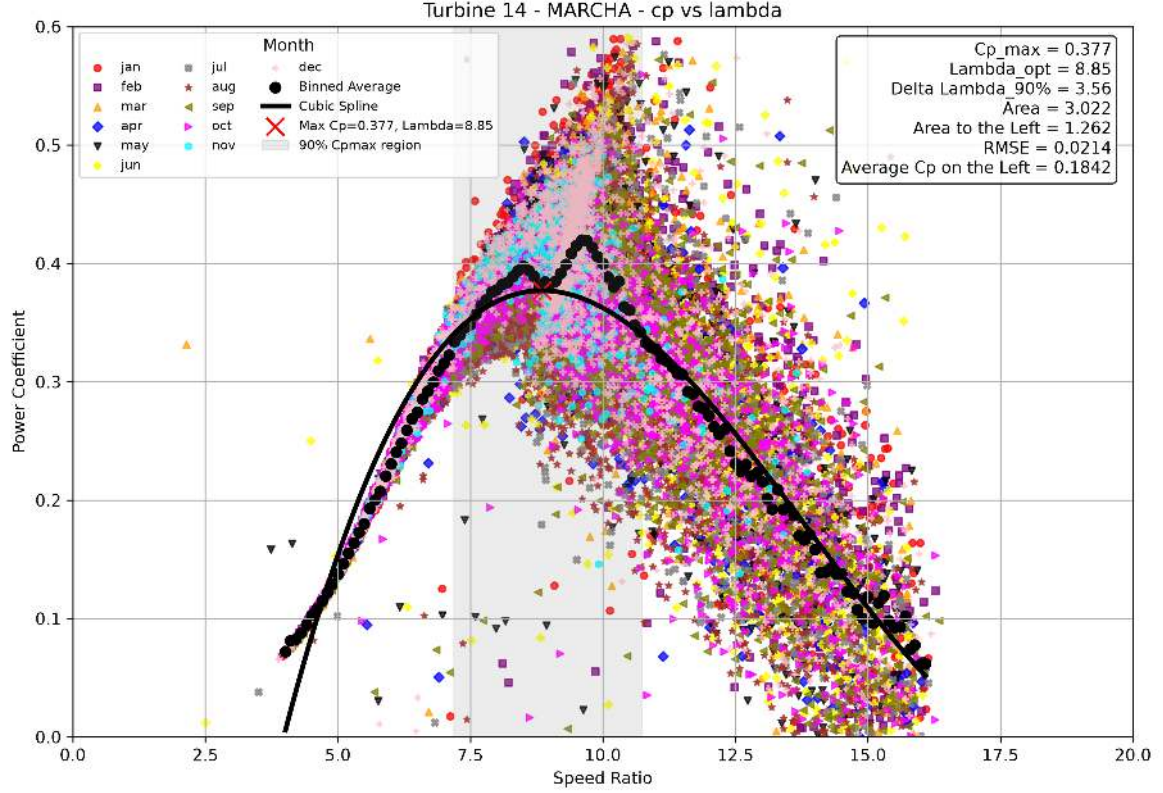


Figure 3: Turbine 14 global C_p -lambda curve during MARCHA operation. The global curve does not clearly isolate the known blade structural damage case.

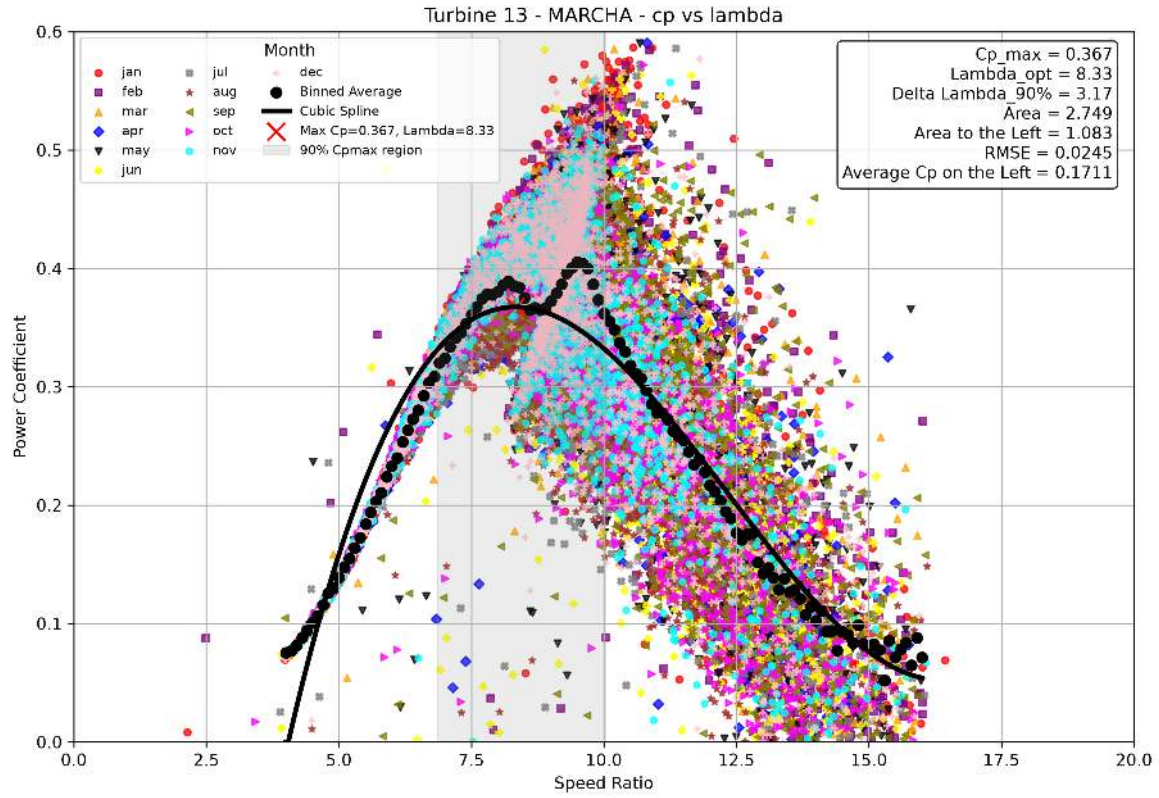


Figure 4: Turbine 13 global Cp-lambda curve during MARCHA operation. The global behavior is similar to Turbine 14, supporting the conclusion that global Cp-lambda is inconclusive.

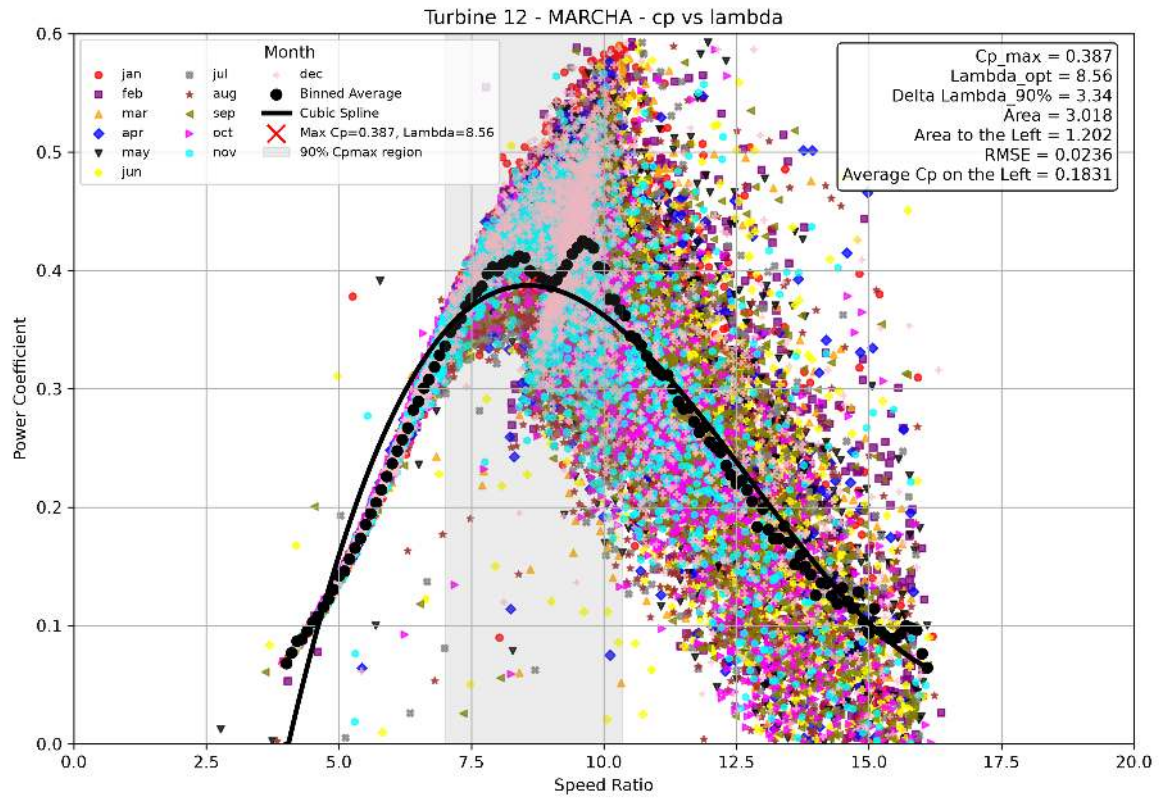


Figure 5: Turbine 12 global Cp-lambda curve during MARCHA operation. Turbine 12 also shows similar global Cp-lambda behavior.

A particularly important contradiction was Turbine 15 in 2021. Turbine 15 had a hydraulic pitch-system problem, but the global C_p -lambda plot showed high apparent performance. This proves that high C_p -lambda metrics do not automatically mean healthy operation.

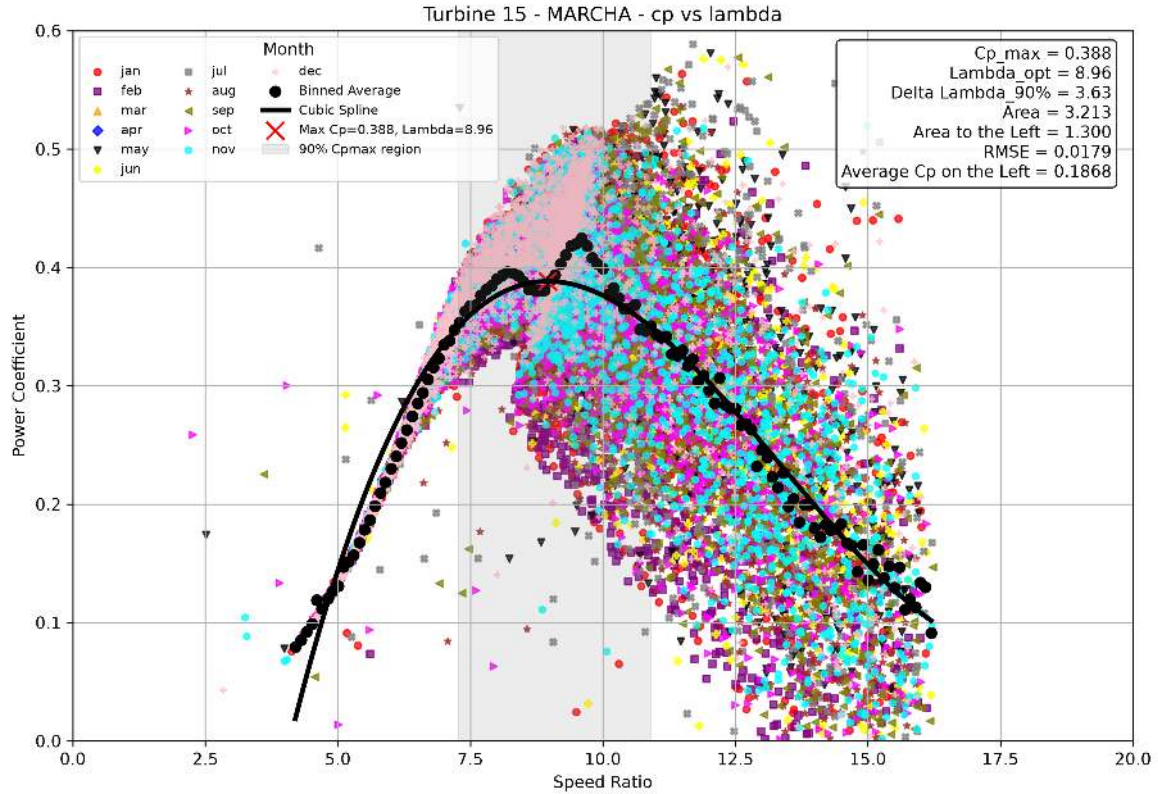


Figure 6: Turbine 15 global C_p -lambda curve during a year with known pitch-system problems. The turbine appears to have high global performance, highlighting that global C_p -lambda curves are not fault-specific.

5 Pitch Angle as the Missing Variable

5.1 Why the Global C_p -Lambda Curve Was Misleading

The central misunderstanding was treating the turbine as though C_p depends only on λ . For a pitch-controlled turbine, this is incomplete. Pitch angle is the angle at which the blade is rotated around its longitudinal pitch axis, and changing pitch angle changes the aerodynamic interaction between the blade and the wind.

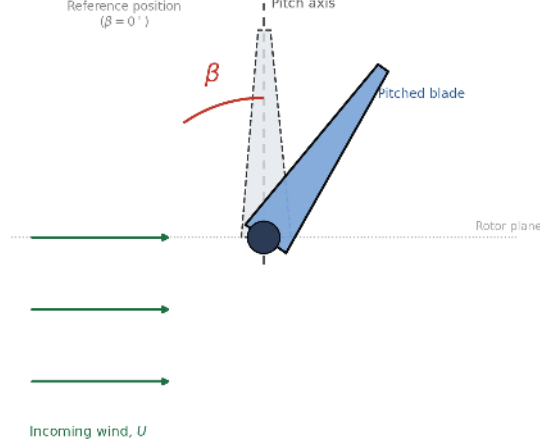


Figure 7: Physical meaning of blade pitch angle. The pitch angle β represents the rotation of the blade around its pitch axis. Changing β changes the aerodynamic orientation of the blade relative to the incoming wind.

The more correct relationship is:

$$C_p = f(\lambda, \beta), \quad (7)$$

where β is blade pitch angle. Technical references and wind turbine performance models use pitch angle and tip-speed ratio as variables affecting the power coefficient [3, 4].

As pitch angle changes, C_p -lambda curves move down and to the left. This explains why the right side of the global C_p -lambda plots has significant variance. The left side tends to behave more consistently, while the right side varies significantly because it contains multiple pitch regimes. Ignoring pitch angle collapses several aerodynamic curves into one global cloud, hiding explanations that can be inferred from pitch-conditioned behavior.

5.2 Pitch-Band C_p -Lambda Analysis

The final analysis therefore splits the data by pitch-angle bands. Instead of one global curve, a separate C_p -lambda curve is plotted for each pitch band. This makes the aerodynamic behavior physically interpretable.

The annual pitch-band figures show that low pitch bands have higher C_p and higher lambda, while higher pitch bands move downward and to the left. This confirms that pitch angle explains much of the scatter in the global C_p -lambda plots.

5.3 Annual Pitch-Band Curves: 2020

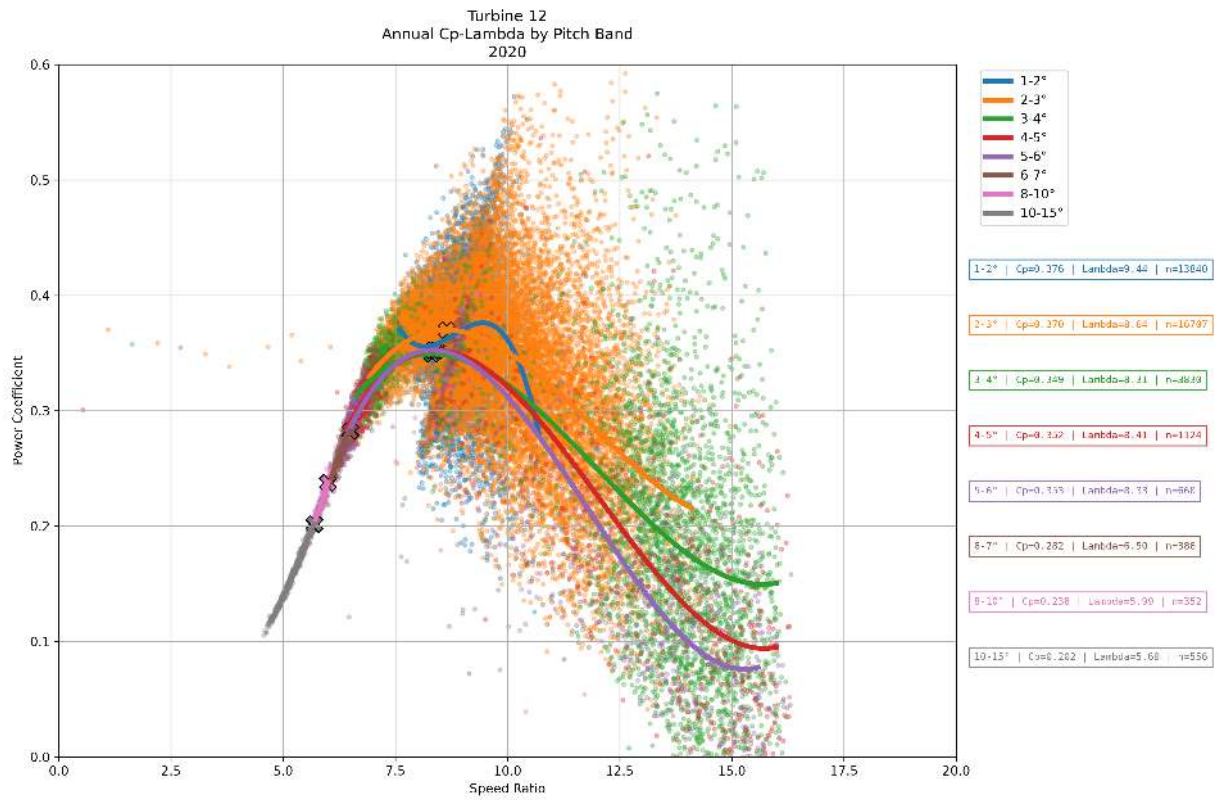


Figure 8: Turbine 12 annual Cp-lambda curves by pitch band in 2020. Separate curves form for each pitch band, showing pitch-conditioned behavior.

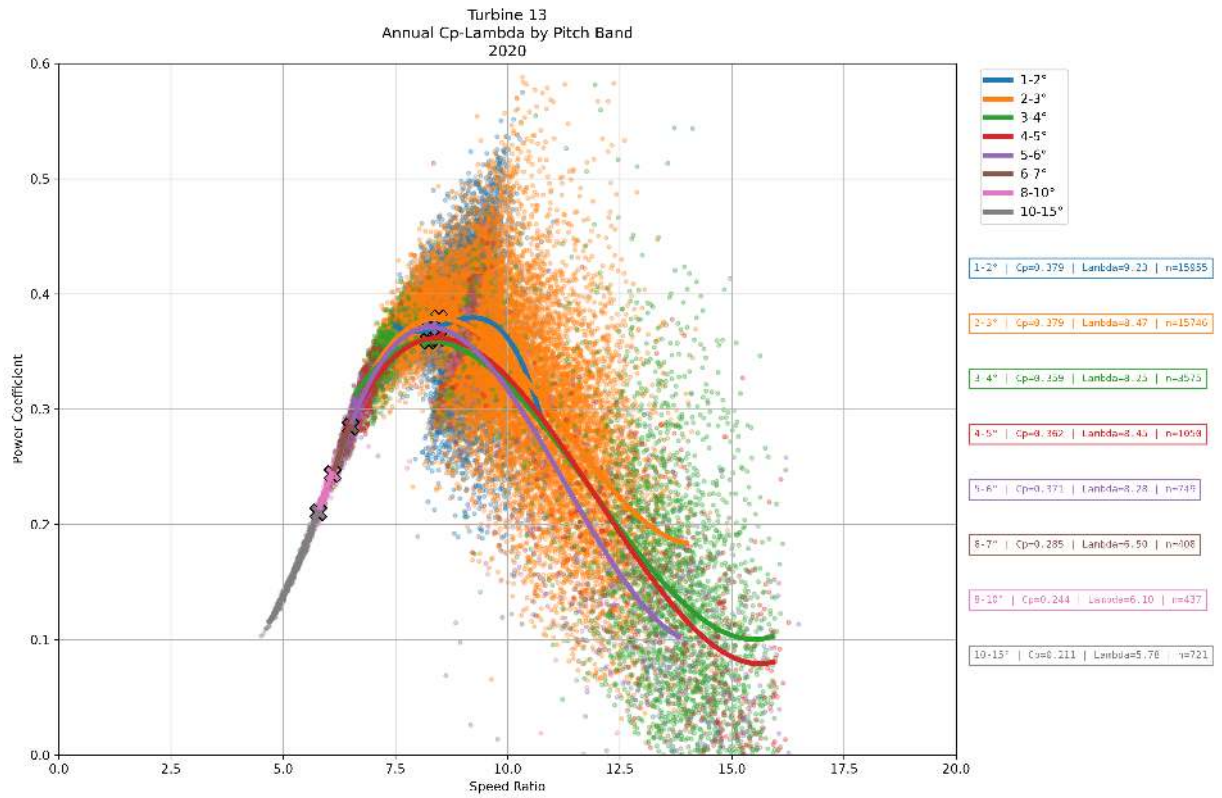


Figure 9: Turbine 13 annual Cp-lambda curves by pitch band in 2020. Turbine 13 shows a physically consistent pitch-family structure.

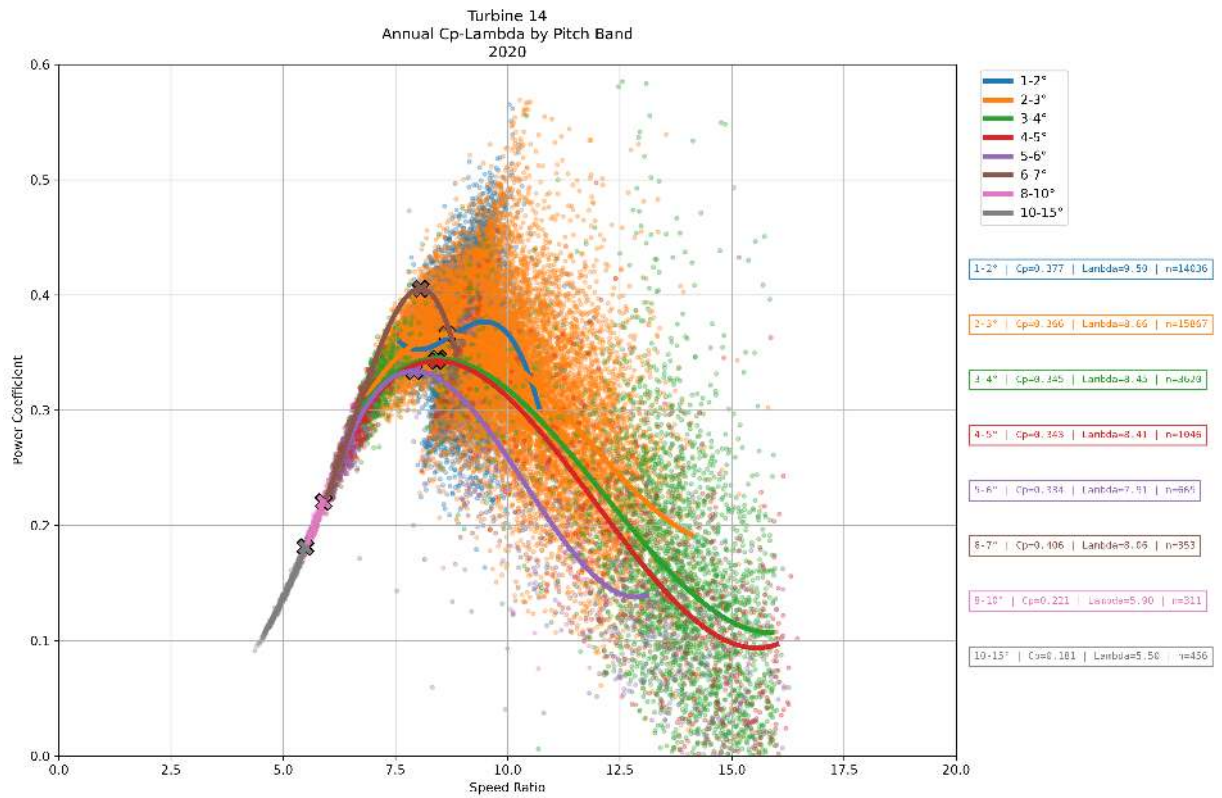


Figure 10: Turbine 14 annual Cp-lambda curves by pitch band in 2020. The low pitch bands are the most populated and stable.

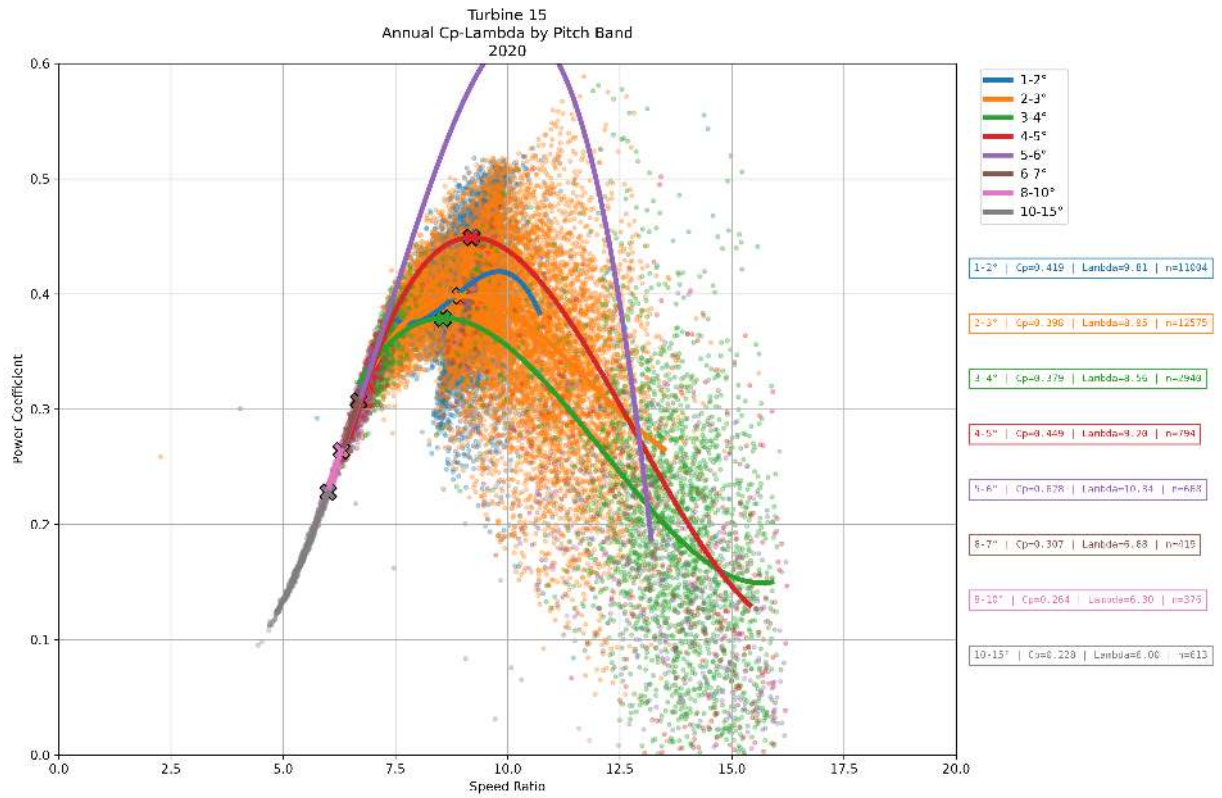


Figure 11: Turbine 15 annual Cp-lambda curves by pitch band in 2020. T15 had a catastrophic blade problem in 2020, and the figure also shows that some pitch-band metrics can be non-physical or spline-sensitive.

5.4 Annual Pitch-Band Curves: 2021

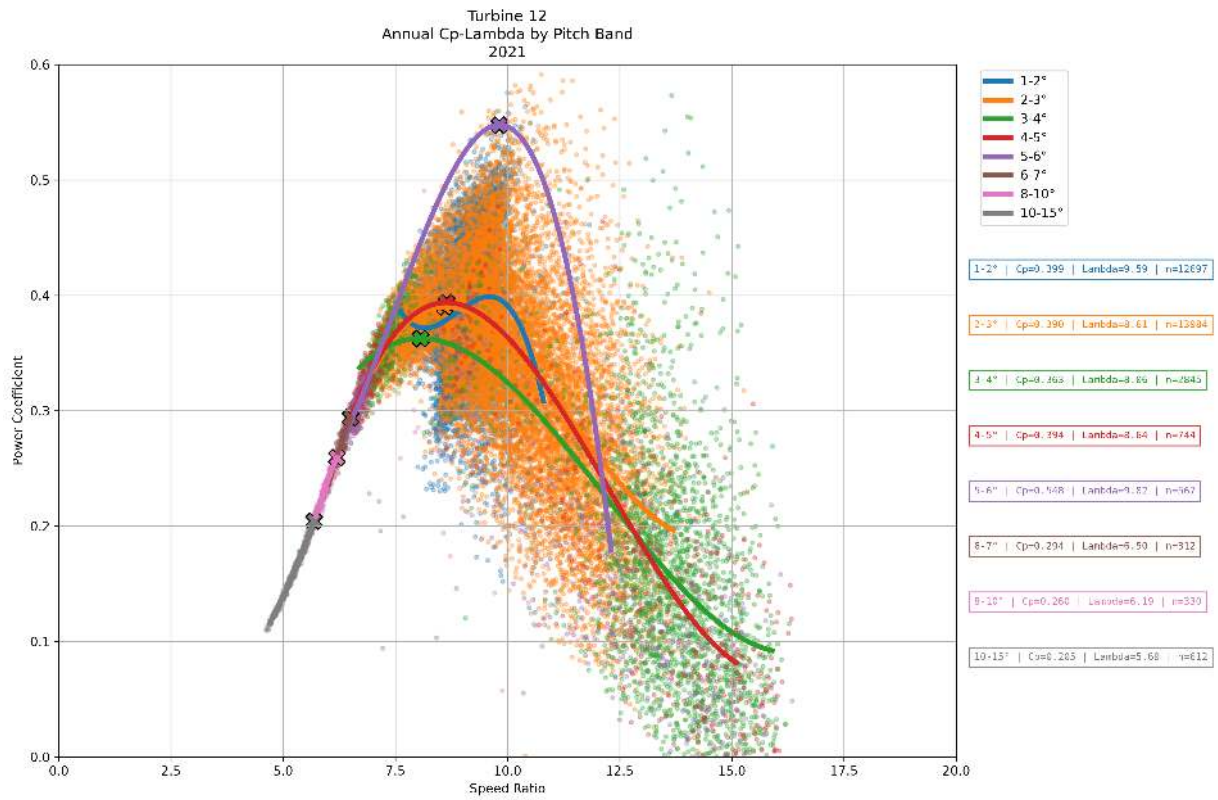


Figure 12: Turbine 12 annual Cp-lambda curves by pitch band in 2021. T12 had pitch-system problems in 2021, so the plot must be interpreted with maintenance context.

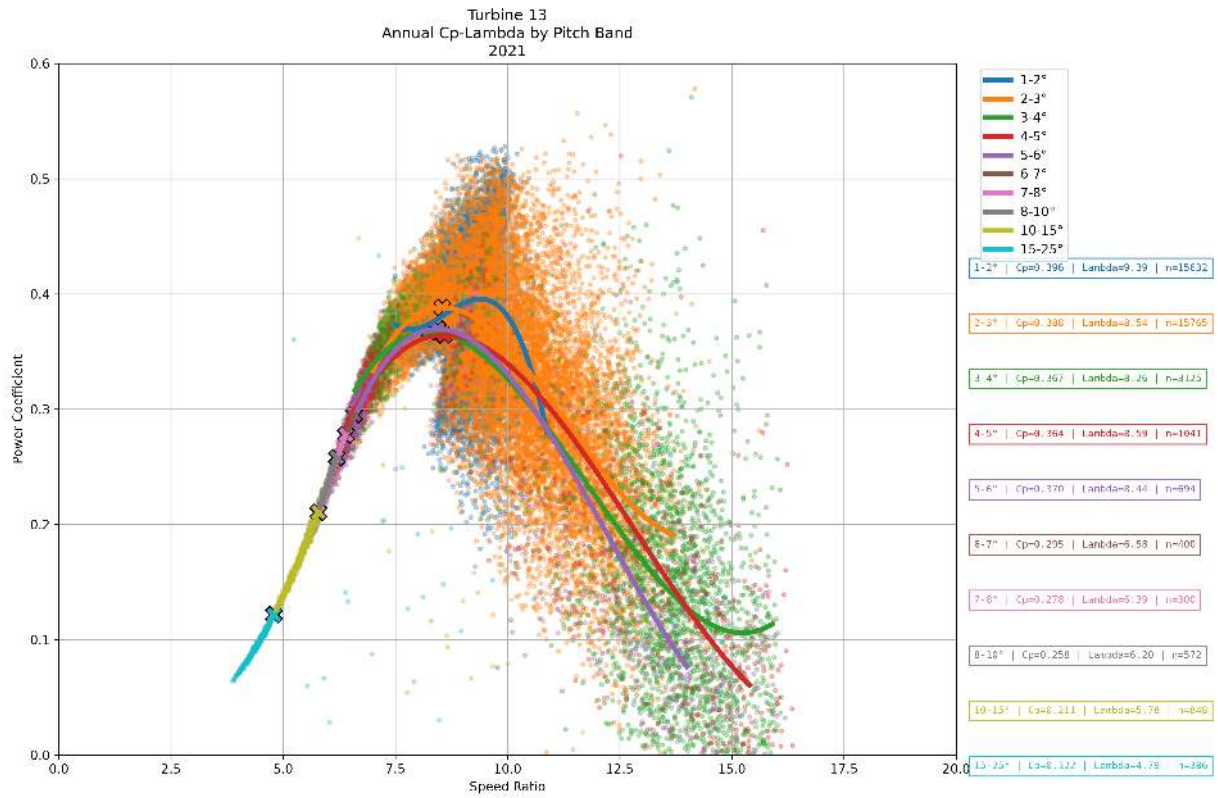


Figure 13: Turbine 13 annual Cp-lambda curves by pitch band in 2021. T13 remains a stable reference case.

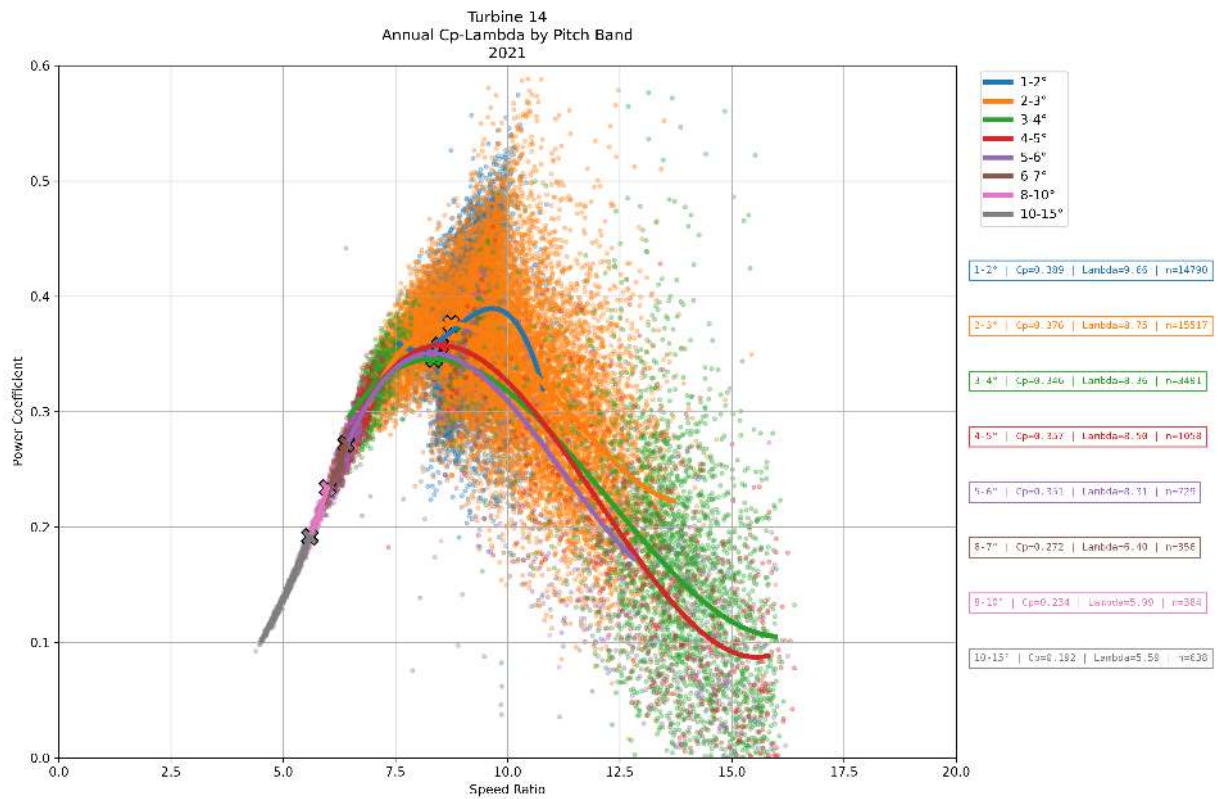


Figure 14: Turbine 14 annual Cp-lambda curves by pitch band in 2021. The curves show expected down-left movement with increasing pitch.

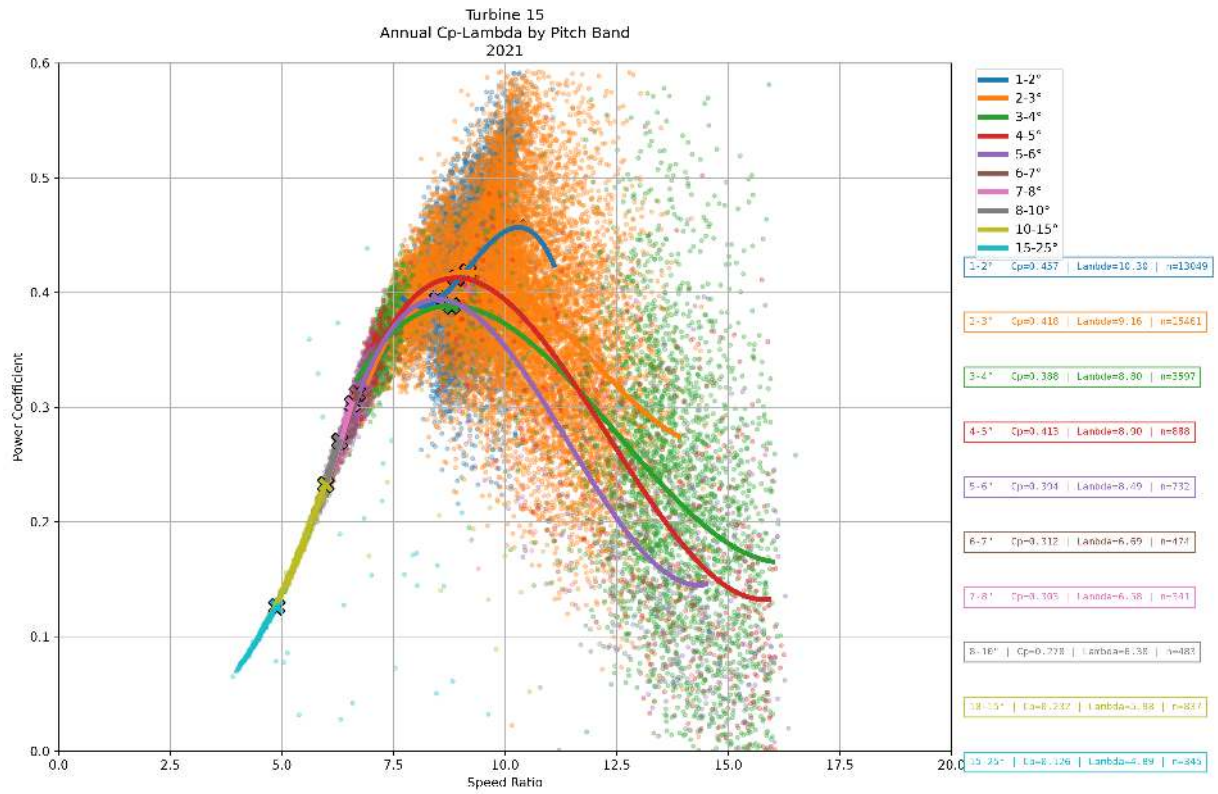


Figure 15: Turbine 15 annual Cp-lambda curves by pitch band in 2021. Despite a known hydraulic pitch-system problem, T15 shows high apparent Cp-lambda performance, proving that high apparent Cp is not automatically healthy operation.

5.5 Annual Pitch-Band Curves: 2022

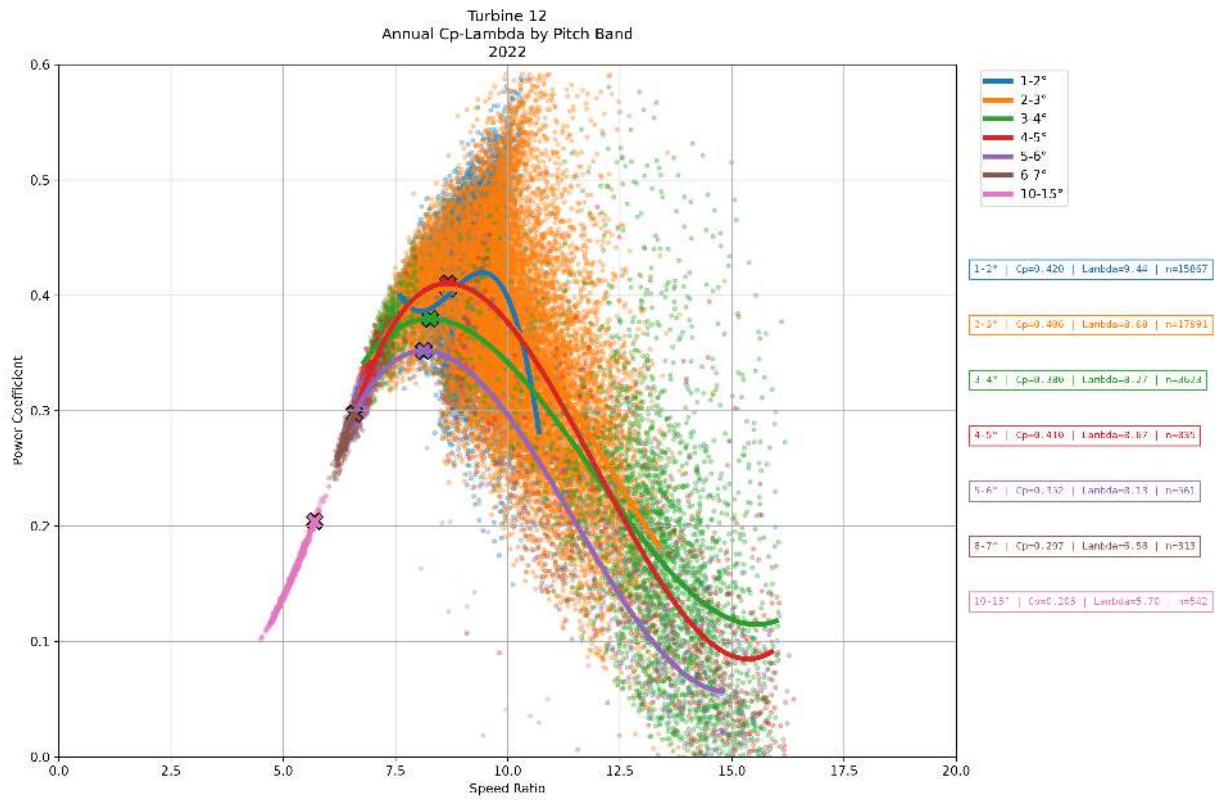


Figure 16: Turbine 12 annual Cp-lambda curves by pitch band in 2022. Several low pitch bands show strong Cp performance.

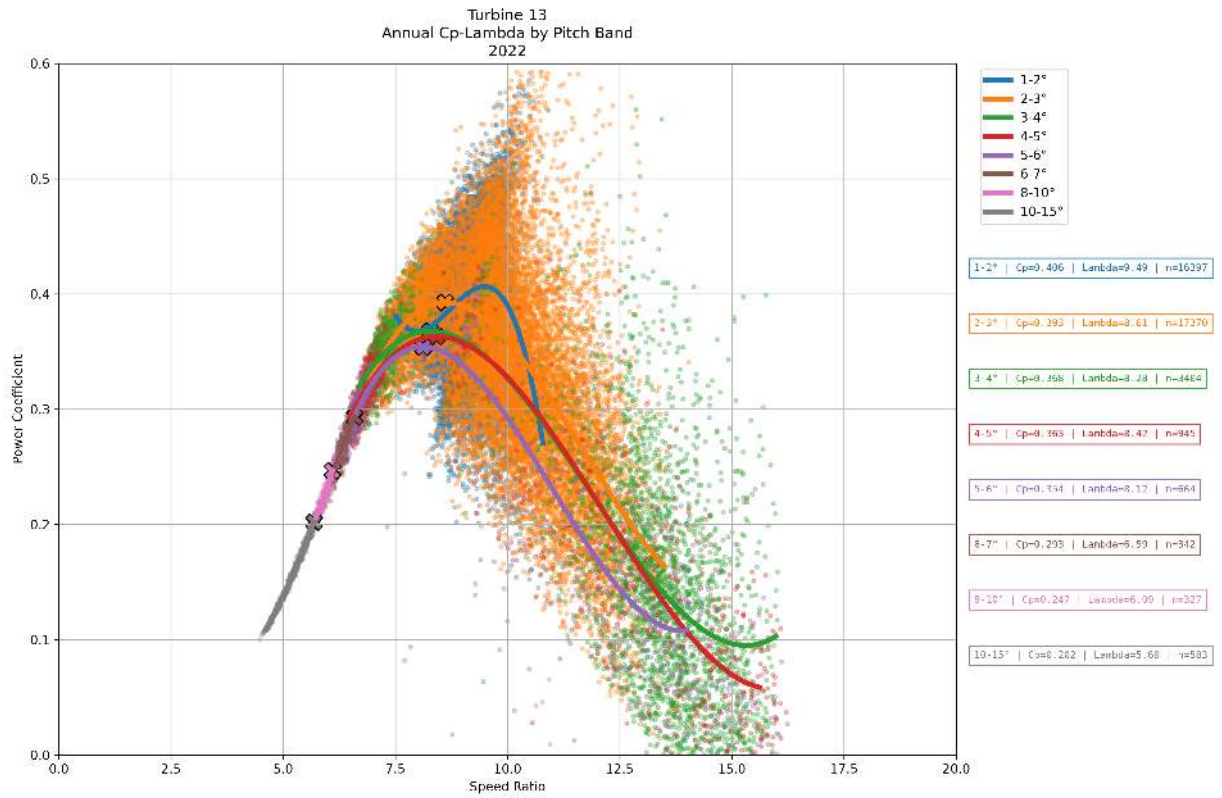


Figure 17: Turbine 13 annual Cp-lambda curves by pitch band in 2022. This is one of the clearest stable pitch-conditioned reference examples.

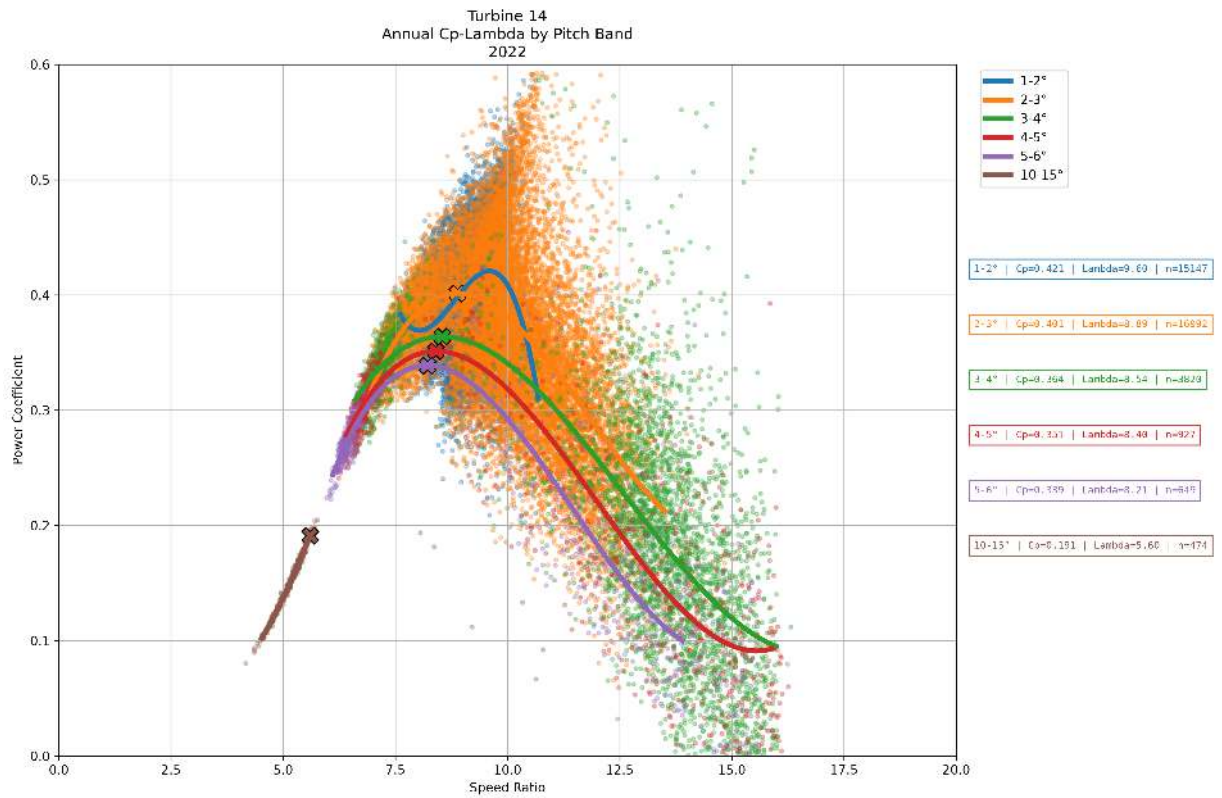


Figure 18: Turbine 14 annual Cp-lambda curves by pitch band in 2022. T14 does not show a clear aerodynamic collapse one year before the known blade structural damage detection.

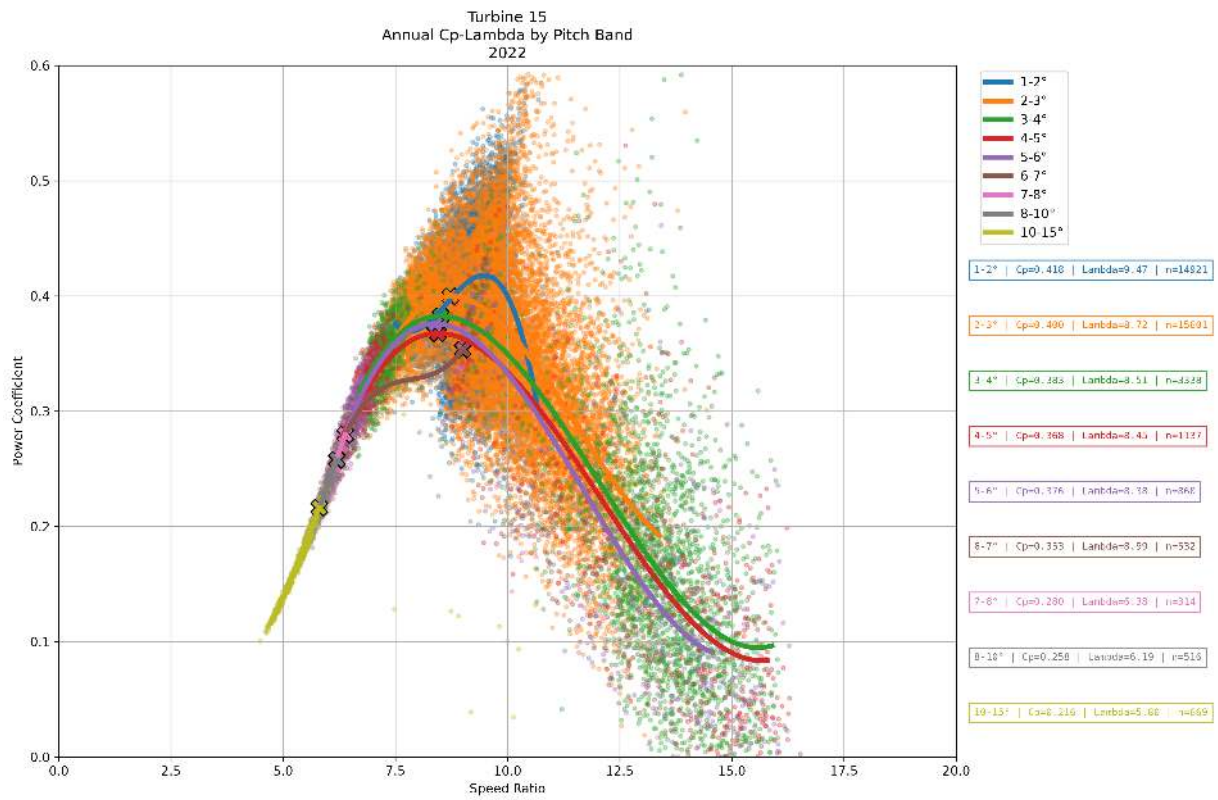


Figure 19: Turbine 15 annual Cp-lambda curves by pitch band in 2022. The behavior is more physically consistent than the preceding problematic years.

5.6 Annual Pitch-Band Curves: 2023

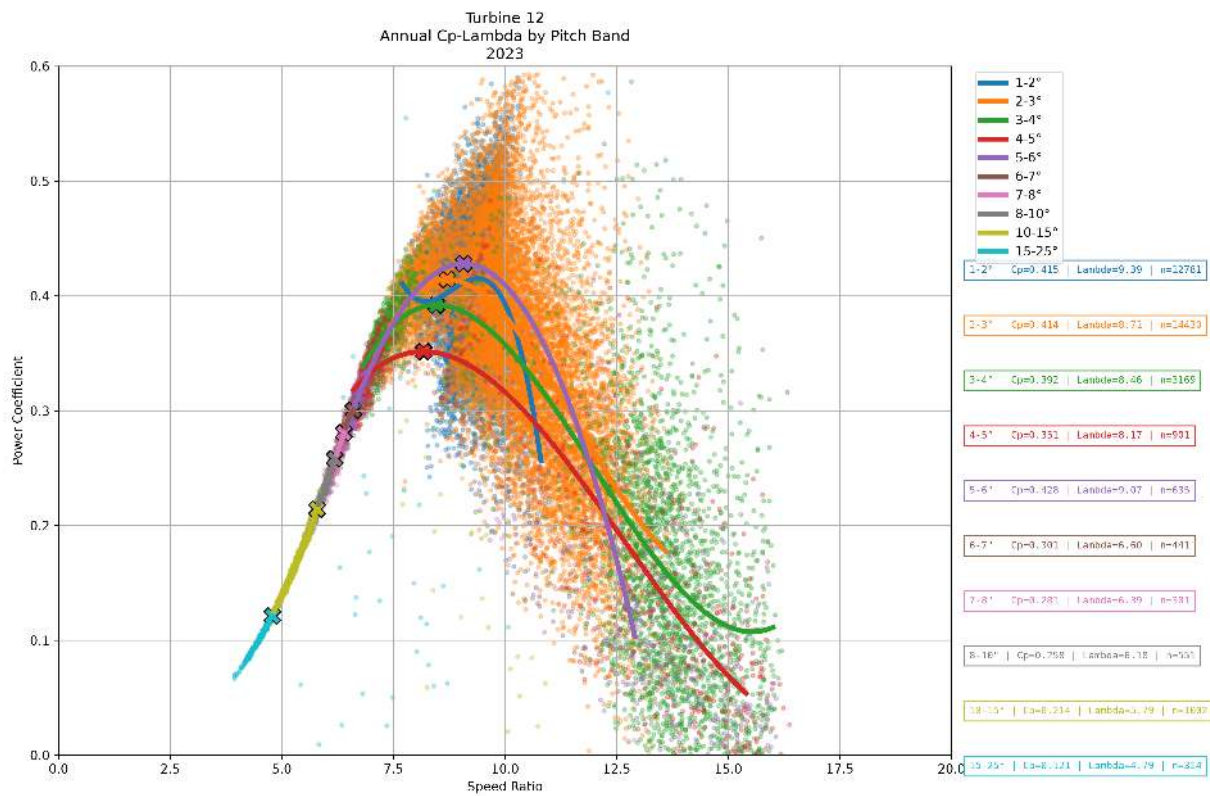


Figure 20: Turbine 12 annual Cp-lambda curves by pitch band in 2023. T12 had pitch-system problems in 2023, so it is not a perfectly healthy baseline in this year.

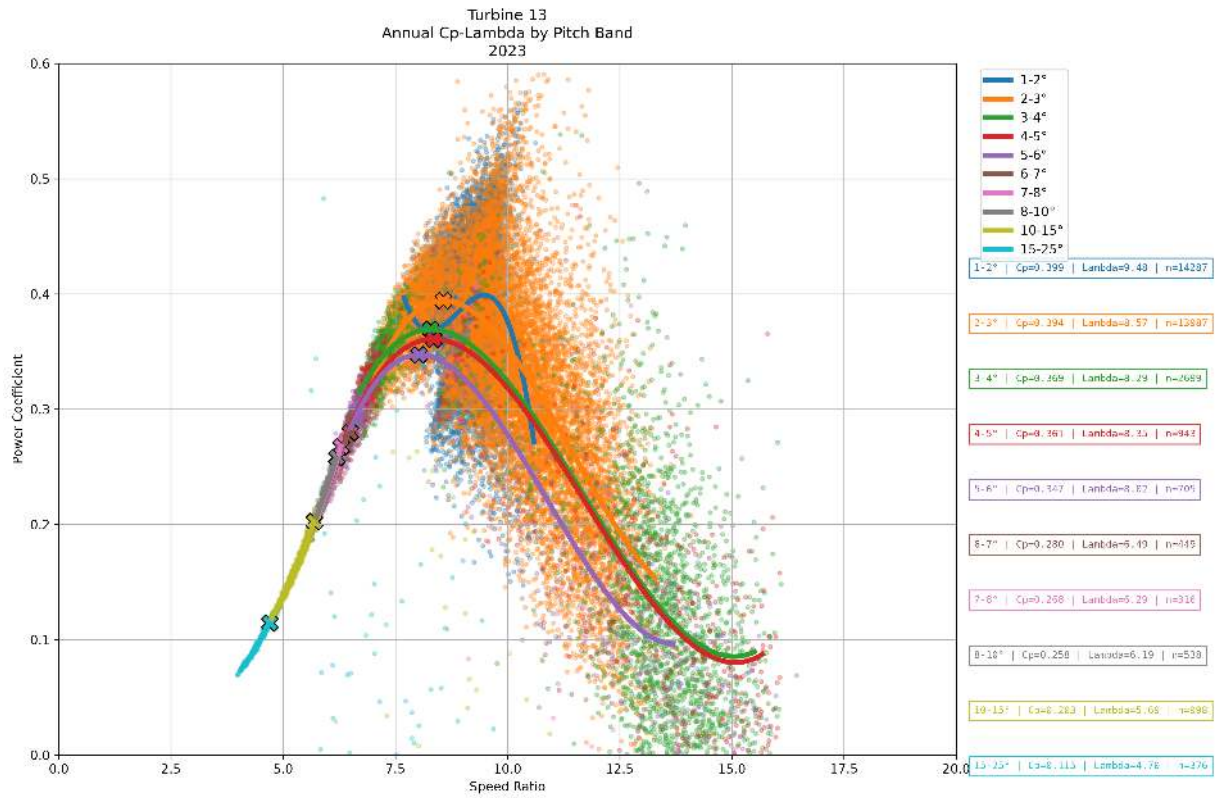


Figure 21: Turbine 13 annual Cp-lambda curves by pitch band in 2023. T13 remains the most stable reference during the target year.

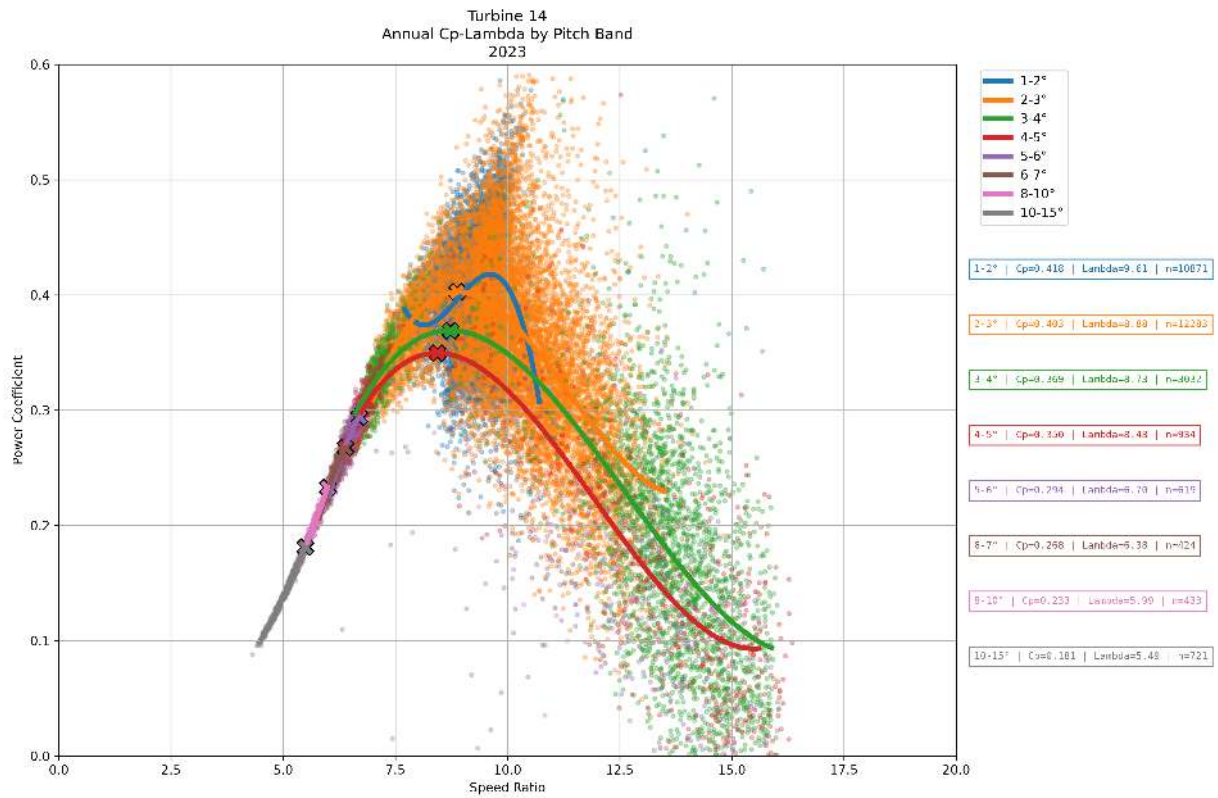


Figure 22: Turbine 14 annual Cp-lambda curves by pitch band in 2023. In the trusted bands 1-2 and 2-3 degrees, T14 does not show a collapse in Cp. The 5-6 degree band is treated as unreliable due to spline instability.

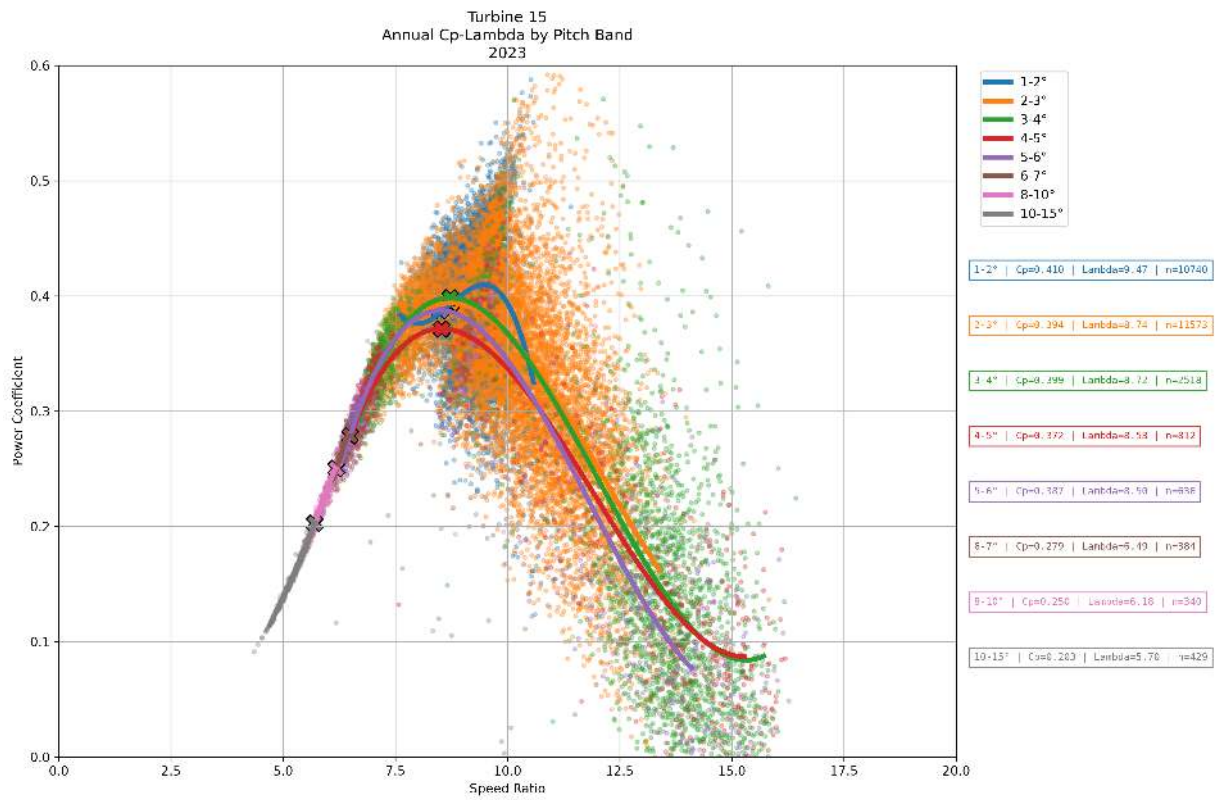


Figure 23: Turbine 15 annual Cp-lambda curves by pitch band in 2023. T15 provides an additional peer comparison for the target year.

5.7 Annual Pitch-Band Curves: 2024

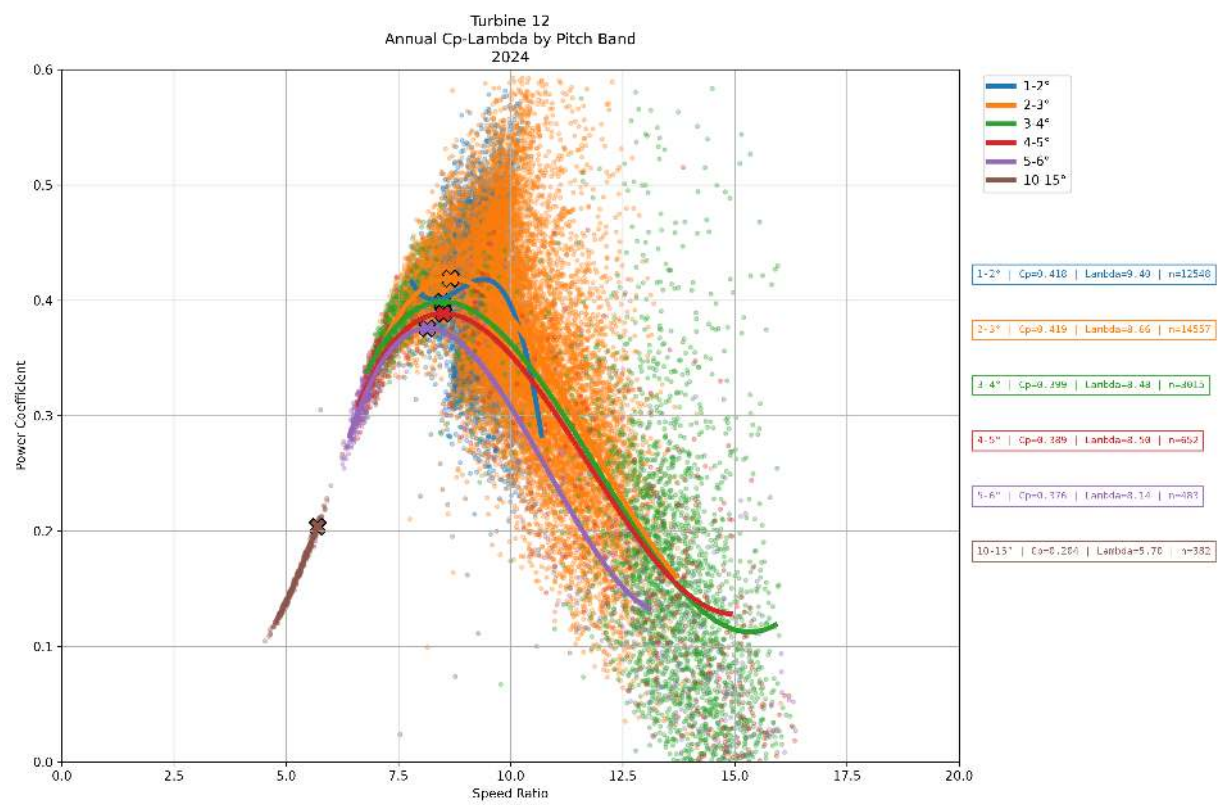


Figure 24: Turbine 12 annual Cp-lambda curves by pitch band in 2024. The low-pitch bands remain strong.

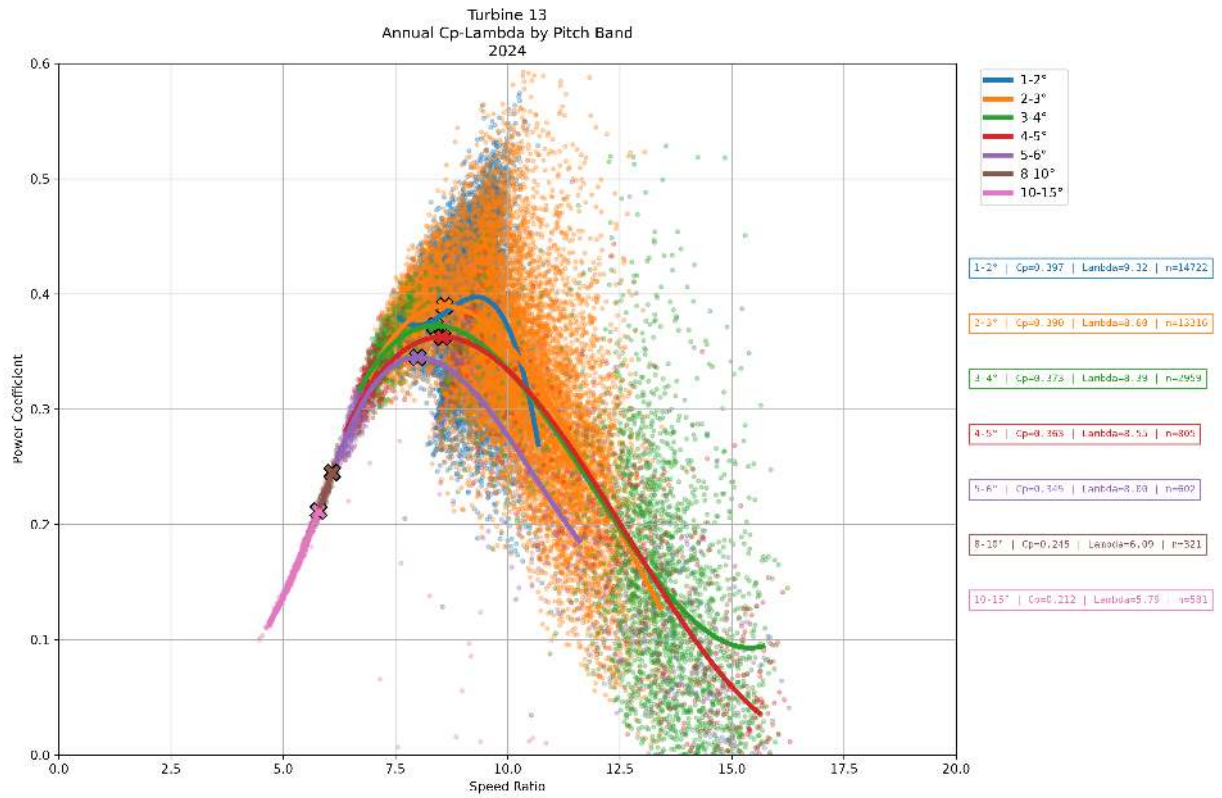


Figure 25: Turbine 13 annual Cp-lambda curves by pitch band in 2024. T13 continues to behave as a stable reference turbine.

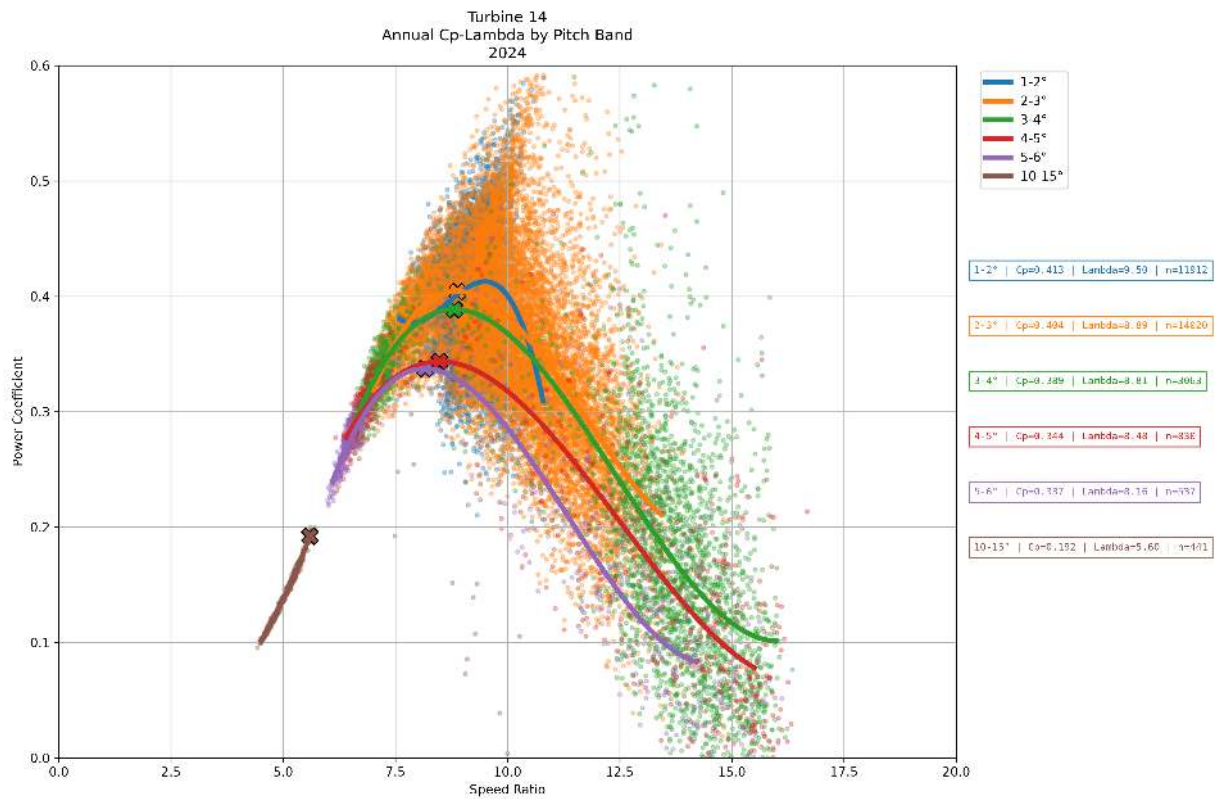


Figure 26: Turbine 14 annual Cp-lambda curves by pitch band in 2024. T14 remains strong in the trusted pitch bands and the unreliable 5-6 degree anomaly does not persist.

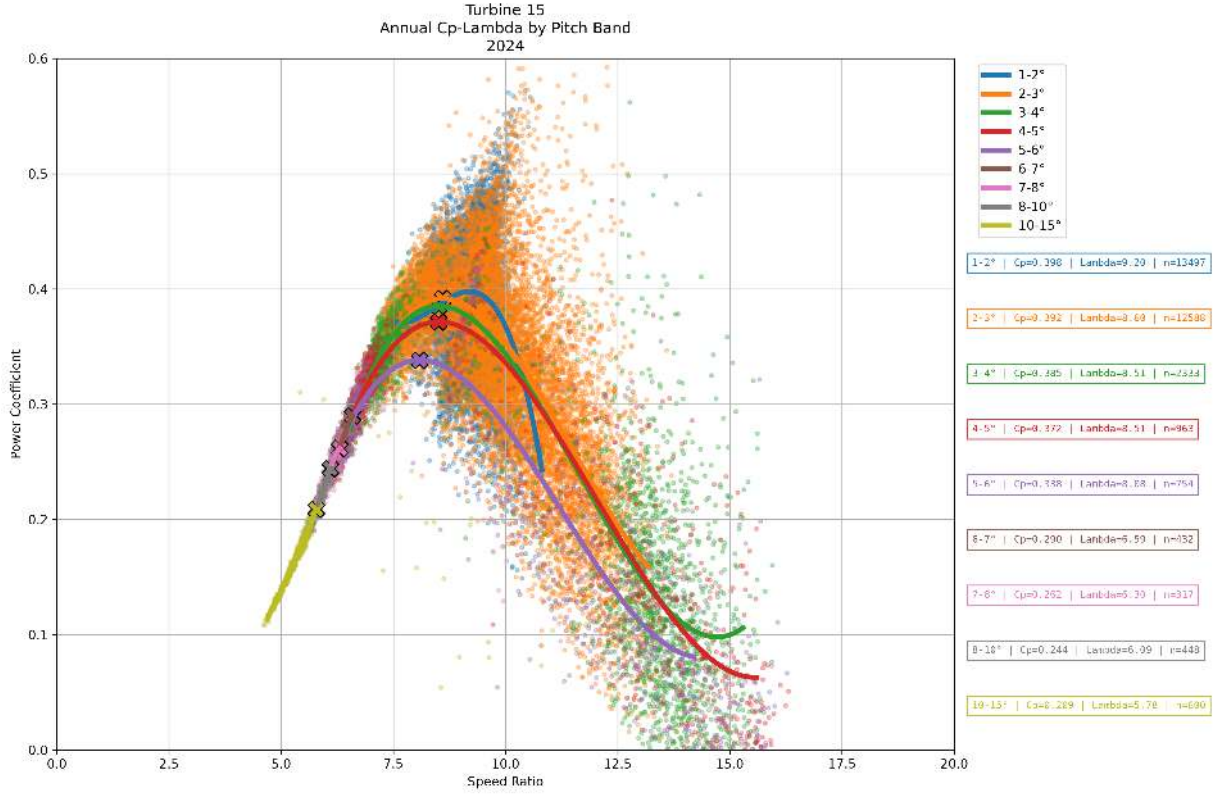


Figure 27: Turbine 15 annual Cp-lambda curves by pitch band in 2024. The turbine shows a physically consistent pitch-family structure.

6 Trusted Pitch Bands and Metric Interpretation

6.1 Why Focus on 1-2 and 2-3 Degrees

The pitch-band analysis shows that not all pitch ranges are equally reliable. Bands such as 5-6 degrees and above can be affected by spline instability, fewer points, different control regimes, and non-physical curve fits. The 5-6 degree behavior for Turbine 14 in 2023 is excluded from the degradation interpretation because it is visually and numerically consistent with a spline problem rather than a stable aerodynamic damage mode.

The most reliable pitch bands are therefore 1-2 degrees and 2-3 degrees. These bands have the highest number of points, the clearest spline behavior, and the most stable aerodynamic production behavior.

6.2 Cp Max in the Trusted Bands

In the trusted pitch bands, Turbine 14 does not show a clear collapse in maximum power coefficient. In pitch band 1-2 degrees, Turbine 14 improves from 2020 to 2022, stays high in 2023, and remains high in 2024. In pitch band 2-3 degrees, Turbine 14 also improves from 2020 to 2022 and stays stable through 2023 and 2024. Therefore, Cp Max by itself does not detect the known blade structural damage.

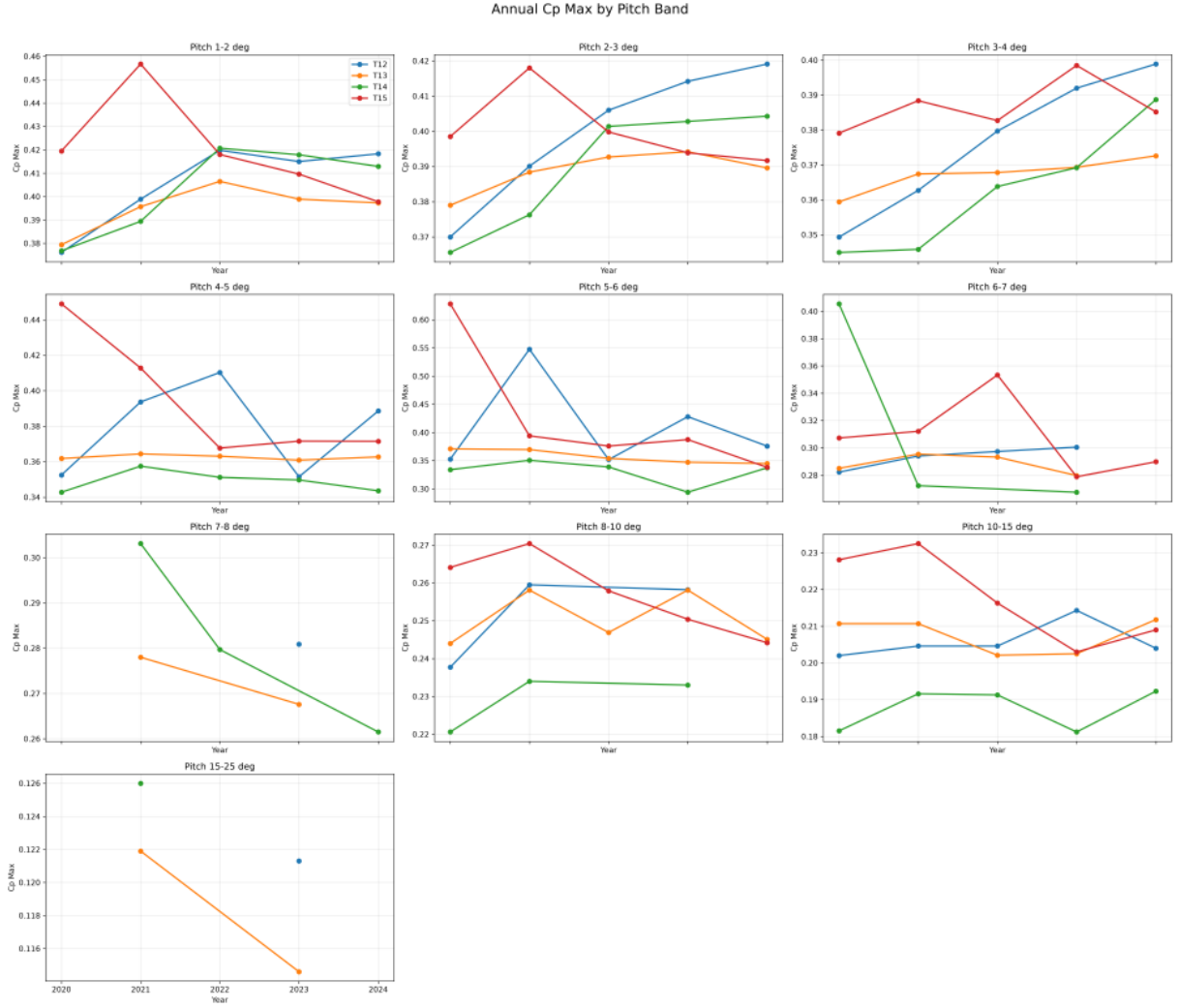


Figure 28: Annual Cp Max by pitch band for Turbines 12, 13, 14, and 15. In the trusted bands 1-2 and 2-3 degrees, Turbine 14 does not show a Cp collapse before or during the 2023 blade structural damage detection.

6.3 Optimum Lambda in the Trusted Bands

Turbine 14 generally has relatively high optimum lambda in the trusted pitch bands. However, this is a stable behavior across the years rather than a sudden 2023 degradation onset. This suggests a turbine-specific operating, control, or tuning characteristic rather than a direct structural damage signal.

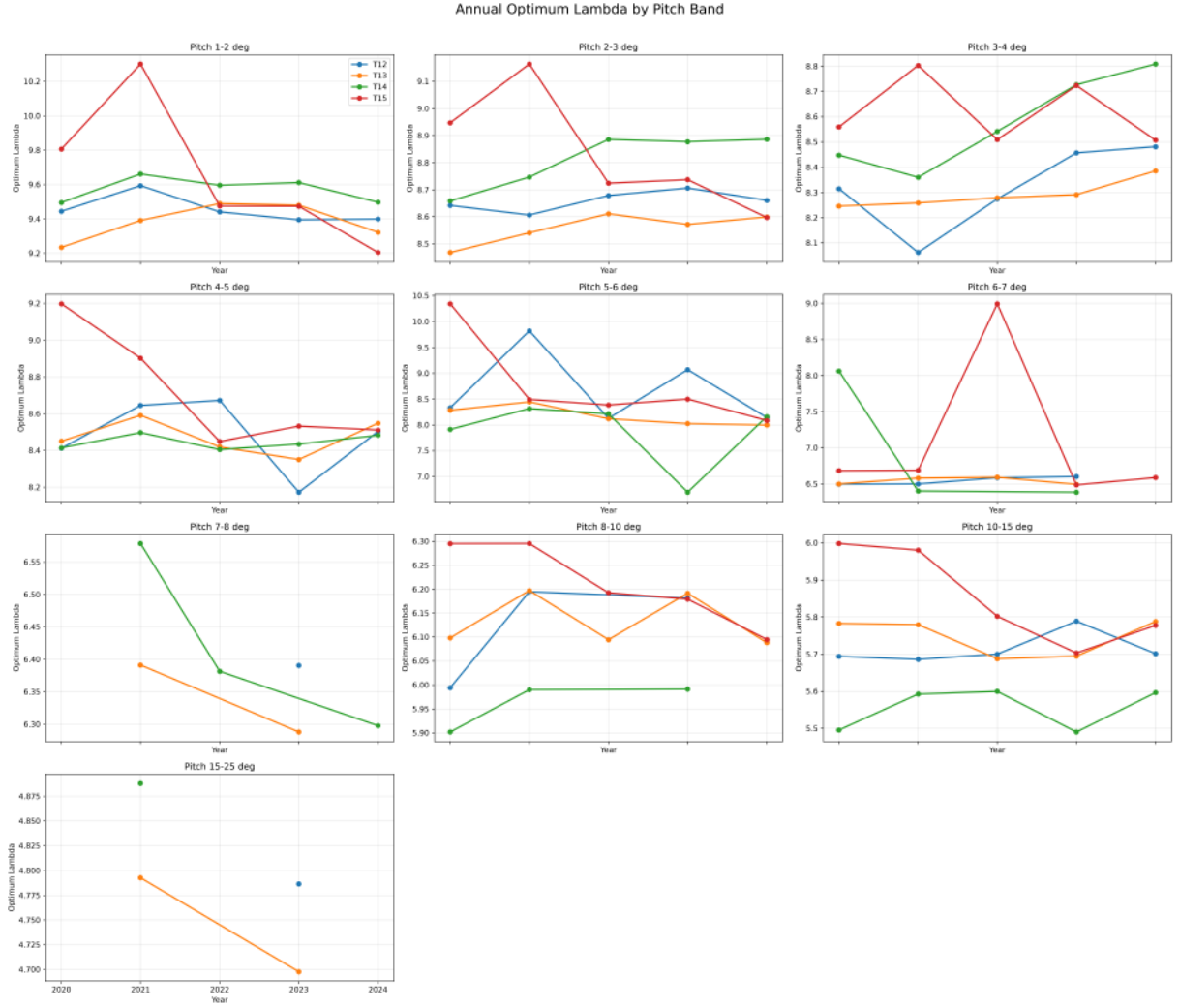


Figure 29: Annual optimum lambda by pitch band for Turbines 12, 13, 14, and 15. Turbine 14 generally has higher lambda in the trusted pitch bands, but this is stable and not a sudden 2023 damage signature.

6.4 Delta Lambda 90 as a High-Efficiency Window Metric

Delta Lambda 90 is the width of the lambda interval where C_p remains above 90 percent of the maximum C_p . It should not only be interpreted as a flattening metric. A larger Delta Lambda 90 means a wider high-efficiency region. A smaller Delta Lambda 90 means a narrower high-efficiency region. Therefore, a decrease can be important because it indicates that the turbine can maintain high C_p over a smaller range of lambda values.

For Turbine 14, the trusted bands show high C_p Max but decreasing Delta Lambda 90. This suggests that the turbine may still reach high peak efficiency, but the range over which it maintains near-maximum efficiency becomes smaller. This can be interpreted as a possible loss of aerodynamic robustness or operating tolerance. However, this remains an aerodynamic observation, not proof of blade structural damage.

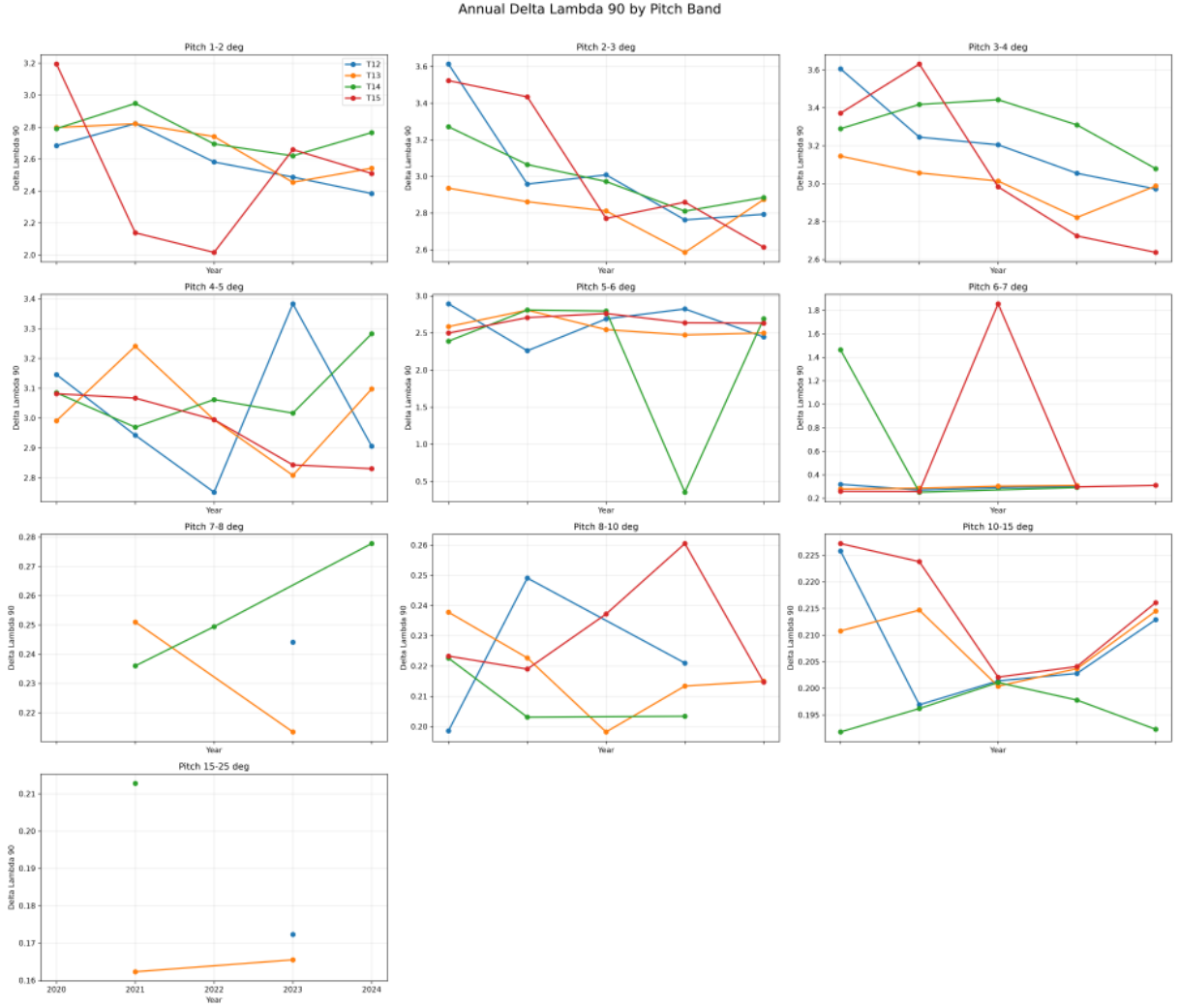


Figure 30: Annual Delta Lambda 90 by pitch band. In the trusted bands, Turbine 14 maintains high Cp but shows a narrowing high-efficiency lambda window, which may indicate reduced aerodynamic robustness rather than a collapse in peak performance.

6.5 Normalized Turbine 14 Overview

The normalized overview is used only to compare relative movement across metrics within each pitch band. It does not preserve physical magnitude. For Turbine 14, the trusted bands show that Cp Max remains high while Delta Lambda 90 decreases. This supports the interpretation that the possible signal is a narrowing high-efficiency operating window rather than a peak Cp collapse.

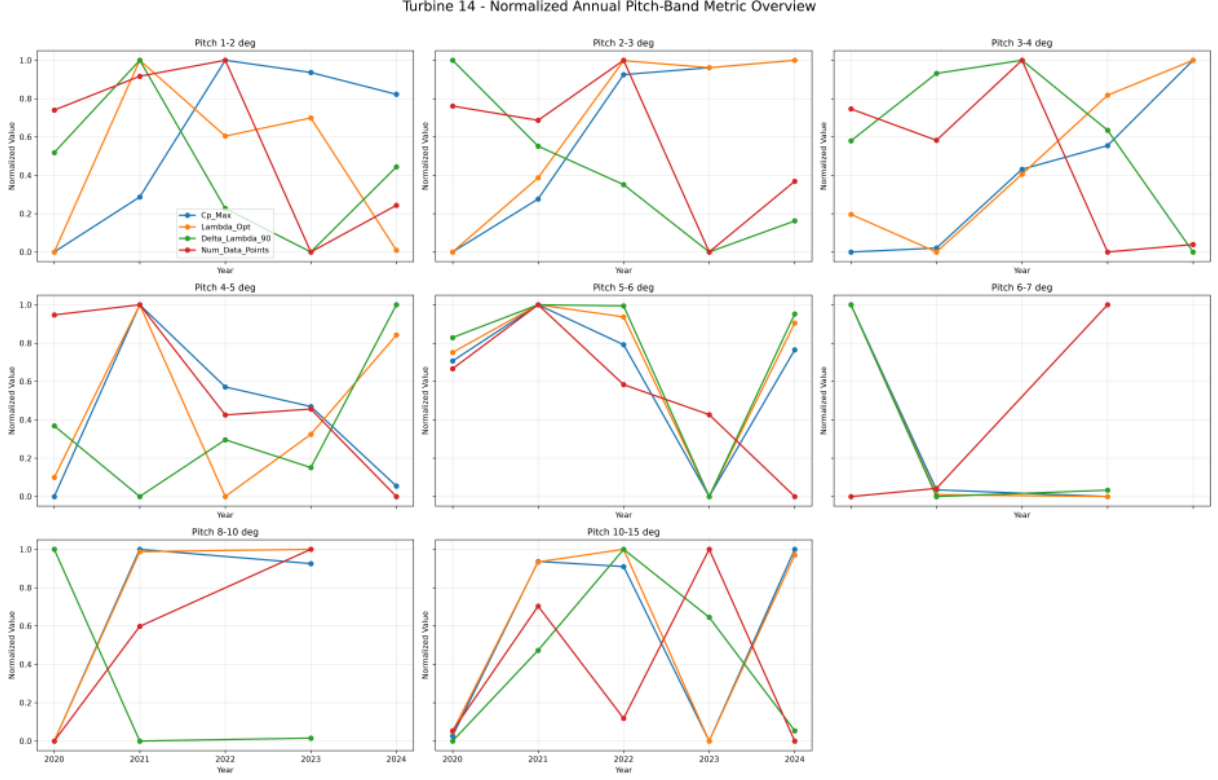


Figure 31: Normalized annual pitch-band metric overview for Turbine 14. The plot is used to compare relative year-to-year movement. In the trusted bands, Cp Max remains high while Delta Lambda 90 narrows.

6.6 Temporal Trend of Delta Lambda 90 in the Trusted Pitch Bands

Sections 6.2–6.5 evaluate each metric largely as a snapshot comparison across turbines within a given year. Reading Figure 30 instead as a *within-turbine, multi-year trend* for Turbine 14, restricted to the two trusted pitch bands identified in Section 6.1, adds an observation not developed elsewhere in this report.

In the **2–3° pitch band**, Turbine 14’s Delta Lambda 90 declines in a clear, nearly monotonic trend from 3.27 in 2020 to 3.07 in 2021, 2.97 in 2022, and 2.81 in 2023, the year the blade structural damage was identified, before recovering to 2.88 in 2024, after the blade was repaired. In the **1–2° pitch band**, the same qualitative pattern is present though less smoothly: Delta Lambda 90 rises from 2.79 in 2020 to a local peak of 2.95 in 2021, falls to 2.69 in 2022 and 2.62 in 2023, and then increases to 2.76 in 2024. In both trusted bands, the series reaches its minimum at the 2023 detection year and is followed by a rebound in 2024.

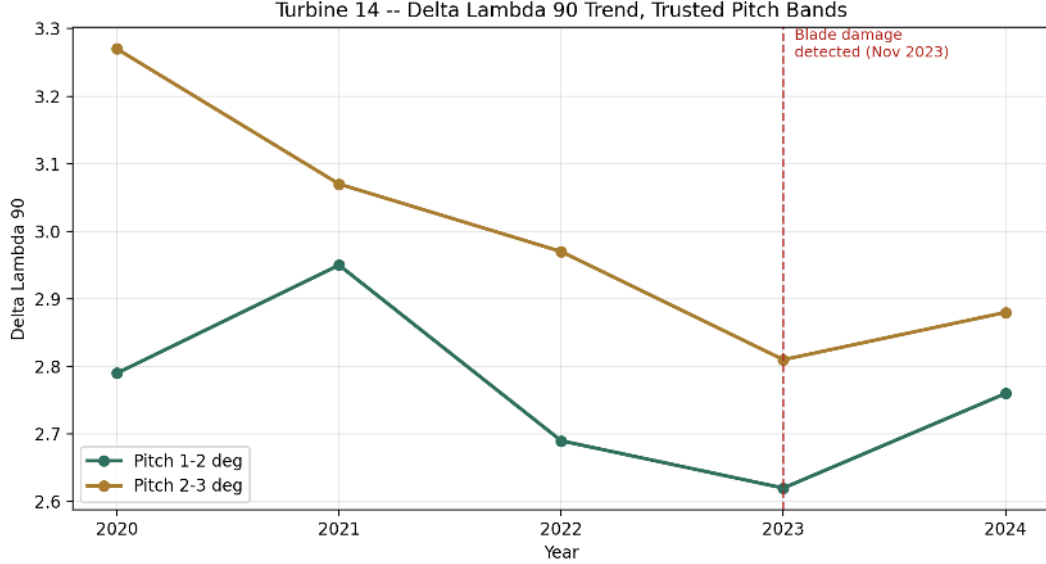


Figure 32: Turbine 14 Delta Lambda 90 by year, trusted pitch bands only (1–2° and 2–3°), extracted from Figure 30. Both bands reach their lowest value in 2023, the detection year, and recover in 2024, after the blade was repaired.

Considered as a single year’s absolute value, Delta Lambda 90 is not sufficient on its own to flag Turbine 14 (Section 6.4), because its level varies turbine-to-turbine and year-to-year for reasons unrelated to blade condition, as seen in the peer comparisons in Figure 30. Evaluated instead as a within-turbine trend over time, in the two trusted pitch bands, the multi-year decline leading into the detection date, followed by a rebound after the repair, is consistent with a physical narrowing of the turbine’s high-efficiency operating window in the lead-up to the blade damage event, and a restoration of that window once the blade was fixed. This trend-based reading therefore does offer usable evidence that blade structural damage can be detected through this metric, ahead of the point where damage becomes severe, provided it is evaluated as a multi-year trajectory rather than a single-year comparison.

7 Final Conclusion

The investigation started with the objective of detecting blade structural damage using only SCADA data and stoppage or maintenance information. The selected case was Turbine 14, where blade structural damage was detected in November 2023. The analysis showed that a conventional power curve based on generated power and wind speed did not distinguish the faulty turbine from the stable reference turbine. Turbine 14 looked similar to Turbine 13 in normal running operation.

Global C_p -lambda analysis added rotor-speed information and was theoretically more aerodynamic, but it also failed to isolate the known fault. Turbine 14 did not clearly stand out from Turbines 12 and 13, and Turbine 15 even showed high apparent C_p -lambda performance during a year with known pitch-system problems. This means global C_p -lambda curves are not fault-specific.

The main methodological discovery was that pitch angle is the missing variable. A real pitch-controlled wind turbine should not be interpreted using a single global $C_p = f(\lambda)$ curve. The correct view is $C_p = f(\lambda, \beta)$. Splitting the data by pitch bands explained the large right-hand variability in the global C_p -lambda plots and revealed distinct aerodynamic curve families.

However, after focusing only on the reliable pitch bands, especially 1-2 and 2-3 degrees, Turbine 14 did not show a clear collapse in C_p Max before or during the 2023 structural blade damage event. Turbine 14 maintained high C_p Max and stable high optimum lambda. The most meaningful

remaining signal is a reduction in Delta Lambda 90, meaning that the high-efficiency operating window may be narrowing even while peak C_p remains high.

Section 6.6 extends this observation by tracking Delta Lambda 90 as a multi-year trend rather than a single-year snapshot. In both trusted pitch bands, the metric declines steadily into 2023, the detection year, and recovers in 2024, after the blade was repaired. Unlike the single-snapshot reading above, this trend-based evidence does support the conclusion that blade structural damage can be detected through this metric, ahead of the point where damage becomes severe, when it is evaluated as a trajectory over multiple years rather than compared at one point in time.

Therefore, the final conclusion is:

Pitch-conditioned C_p -lambda analysis improves the physical interpretation of SCADA data and can reveal aerodynamic operating differences hidden in global plots. A single year's Delta Lambda 90 value, compared across turbines, is not sufficient to classify a blade structural damage event. However, tracked as a multi-year trend within the trusted 1-2 and 2-3 degree pitch bands, Delta Lambda 90 does provide usable evidence: for Turbine 14, the metric declines steadily in the years leading into the November 2023 detection date and recovers afterward, once the blade was repaired. SCADA-based aerodynamic curves, read as trends rather than snapshots, must still be combined with maintenance logs, stoppage records, inspection findings, and data-quality checks, but the Delta Lambda 90 trend in the trusted pitch bands is a concrete, usable early-detection signal for this case.

8 Further Studies

8.1 Move from Global Curves to Pitch-Conditioned Diagnostics

Future work should not use global C_p -lambda curves as the main diagnostic tool. The diagnostic should be pitch-conditioned, either by pitch-angle bands or by a two-dimensional aerodynamic surface:

$$C_p = f(\lambda, \beta). \quad (8)$$

8.2 Curve Quality Classification

Before interpreting a pitch-band curve, a quality classifier should determine whether the curve is reliable. The quality classifier should use number of points, number of bins, RMSE, lambda coverage, spline shape, and physical validity such as $C_p \leq 0.593$. Curves that fail these checks should be labeled as data-quality unreliable, not as degradation.

8.3 AI Classification by Pitch Range

A future classification AI should classify turbine behavior separately for each pitch range. The AI should not use one global turbine score. Instead, it should use features extracted by turbine, year or month, and pitch band. Candidate inputs include C_p Max, Lambda Opt, Delta Lambda 90, RMSE, Num Data Points, Num Bins, Peak Sharpness, Area, peer-median differences, and turbine historical deviations.

The output classes should include:

- Normal behavior.
- Data-quality or spline artifact.

- Aerodynamic anomaly.
- Pitch-system or hydraulic pitch anomaly.
- Possible blade structural damage.
- Possible generator or control anomaly.
- Inconclusive.

8.4 Use Maintenance Labels for Supervised Learning

The known events should be used as supervised labels. Examples include Turbine 15 blade breakage in 2020, Turbine 15 hydraulic pitch-system problem in 2021, Turbine 14 pitch-system problem in 2020, Turbine 14 blade structural damage in 2023, Turbine 12 pitch-system problems in 2021 and 2023, and Turbine 13 as a stable reference. This will allow the model to learn differences between fault types and avoid interpreting all anomalies as blade structural damage.

8.5 Recommended Future Workflow

The recommended workflow is:

1. Filter SCADA data to valid MARCHA operation.
2. Split data by pitch-angle bands.
3. Fit C_p -lambda curves only for reliable pitch bands.
4. Extract C_p Max, Lambda Opt, Delta Lambda 90, RMSE, and data-quality features.
5. Compare each turbine against its own history and nearby peers.
6. Join stoppage and maintenance labels.
7. Train a classification AI model by pitch band.
8. Validate AI predictions against inspection and maintenance records.

The key future objective is not only to detect that a turbine is abnormal, but to classify the most likely type of abnormality and quantify confidence.

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